

MODULE 01 / 40

Northern Cardinal

*Cardinalis cardinalis***QUICK STATS**

Length	8.3 – 9.3 in (21 – 23.5 cm)
Wingspan	9.8 – 12.2 in (25 – 31 cm)
Weight	1.5 – 1.7 oz (42 – 48 g)
Lifespan	Up to 15 years (avg. 3 in wild)

PRESENTATION

The Northern Cardinal is one of the most iconic and instantly recognizable birds visiting North American gardens year-round. The brilliant scarlet male, with his dramatic crest and orange-red bill, is among the most photographed birds on the continent. The female, dressed in warm buff-brown with reddish tinges on her crest, wings and tail, is no less elegant. Both sexes remain on their territory throughout winter, bringing color and song to snow-covered gardens when most other species have migrated south.

HABITAT & RANGE

The Northern Cardinal thrives across the eastern half of North America, from southern Maine and the Great Lakes south through Florida, and westward to the Great Plains of Texas and Oklahoma. Populations are also well-established in Arizona, New Mexico, and parts of California. Southern Ontario and Quebec host significant breeding populations. The species has expanded its range northward over the past century, likely aided by the proliferation of bird feeders and milder winters. It favors woodland edges, dense shrubby thickets, hedgerows, suburban gardens, and parks with mature trees and dense understory cover.

DIET & FEEDING

- Black-oil sunflower seeds — the undisputed favorite
- Safflower seeds — especially attractive to cardinals
- White millet and cracked corn
- Wild berries: dogwood, holly, elderberry, sumac
- Insects and larvae (essential protein during breeding season)
- Fruit: grapes, mulberries, raspberries

BEHAVIOR & SOCIAL LIFE

Cardinals are territorial songbirds that maintain year-round home ranges of roughly 2–4 acres. Males fiercely defend their territory through song and, famously, by attacking their own reflection in windows — mistaking it for a rival. They are monogamous and pairs often remain together through multiple breeding seasons. Outside the breeding season, cardinals may gather in loose flocks of up to 70 individuals at abundant food sources. Males frequently feed females during courtship — a behavior called mate-feeding — passing seeds beak-to-beak.

NESTING & BREEDING

Breeding season runs from March through September, with females typically raising 2 to 4 broods per year — one of the highest rates among North American songbirds. The female constructs a cup-shaped nest over 3–9 days using twigs, bark strips, grasses and leaves, lined with fine grasses and hair. Nests are built 1–15 feet above ground in dense shrubs, vines or low trees. Each clutch contains 2–5 eggs (usually 3), pale whitish or greenish with brown speckles. Incubation lasts 11–13 days, performed almost exclusively by the female. Nestlings fledge at 9–11 days and the male takes over their care while the female begins a new nest.

SONG & COMMUNICATION

The Northern Cardinal is renowned for its rich, loud whistled song — a series of clear slurred notes often described as "cheer-cheer-cheer" or "birdy-birdy-birdy." Both males and females sing, which is unusual among North American songbirds. The female often sings from the nest, possibly communicating food needs to her mate. Cardinals have a repertoire of 16 or more distinct song patterns. Their sharp metallic "chip" call is an alarm note used to warn of predators. Song peaks in late winter and early spring as males establish territories.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable / Increasing
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With an estimated population of over 100 million individuals, the Northern Cardinal is one of the most abundant songbirds in North America. Its range has expanded northward significantly since the 1900s. It is the official state bird of 7 US states — more than any other species.

FEEDER TIPS

- Best feeder type: hopper feeder or large platform feeder
- Preferred seeds: safflower (squirrels tend to avoid it) and black-oil sunflower
- Cardinals feed most actively at dawn and dusk
- Place feeders near dense shrubs — cardinals prefer cover close by
- Offer a shallow birdbath: cardinals bathe and drink regularly

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Male body	Vivid red — cadmium red
Facial mask	Deep black
Bill	Orange-coral
Female body	Warm sandy brown
Wings & crest ■	Rusty-red highlights
Eye	Dark brown, black ring

COLOR ME!

Northern Cardinal • *Cardinalis cardinalis* • Module 01



MODULE 02 / 40

Black-capped Chickadee

*Poecile atricapillus***QUICK STATS**

Length	4.7 – 5.9 in (12 – 15 cm)
Wingspan	6.3 – 8.3 in (16 – 21 cm)
Weight	0.3 – 0.5 oz (9 – 14 g)
Lifespan	Up to 12 years (avg. 2–3 in wild)

PRESENTATION

The Black-capped Chickadee is one of the most beloved and recognizable backyard birds across northern North America. Bold, curious, and remarkably tolerant of humans, it is often the first species to discover a newly installed feeder. Its cheerful "chick-a-dee-dee-dee" call is one of the most familiar sounds of North American winters. Despite its tiny size, the chickadee is supremely adapted to cold climates, capable of surviving temperatures well below freezing through a combination of extraordinary memory, metabolic flexibility, and dense winter plumage.

HABITAT & RANGE

The Black-capped Chickadee occupies a vast range across the northern tier of North America, from Alaska and the Yukon east to Newfoundland and Nova Scotia, and south through the northern United States including New England, the Great Lakes region, and the Rocky Mountain states down to northern New Mexico. It inhabits deciduous and mixed forests, forest edges, riparian corridors, suburban parks, and wooded gardens. It is a permanent resident throughout its range and does not migrate, relying instead on cached food and physiological adaptations to endure harsh winters.

DIET & FEEDING

- Black-oil sunflower seeds — the top choice at any feeder
- Suet cakes — critical high-energy winter food source
- Nyjer (thistle) seed and hulled sunflower chips
- Peanut pieces and peanut butter mixtures
- Mealworms (live or dried) during breeding season
- Wild: insects, insect eggs, spiders, berries, seeds

BEHAVIOR & SOCIAL LIFE

Chickadees are highly social birds that form stable winter flocks of 6–12 individuals with a clear dominance hierarchy. These flocks often serve as the nucleus for mixed-species foraging groups, attracting nuthatches, woodpeckers, and warblers. Within the flock, dominant birds feed first and claim the best roost sites. The chickadee's "chick-a-dee" alarm call is remarkably sophisticated: the number of "dee" notes appended to the call signals the level of threat, with more notes indicating a more dangerous predator nearby. Researchers have identified it as one of the most complex vocalizations in the animal kingdom.

NESTING & BREEDING

Breeding pairs form in late winter as flocks begin to break up. Chickadees are cavity nesters, excavating their own nest holes in soft rotting wood or using natural cavities and nest boxes. The female lines the cavity with moss, plant fibers, fur, and feathers, creating a deep insulating cup. Clutches typically contain 6–8 white eggs with reddish-brown speckles. Incubation lasts 12–13 days, performed by the female alone while the male brings her food. Both parents feed the nestlings, which fledge at 16 days. Pairs raise one brood per year.

SONG & COMMUNICATION

The chickadee has two primary vocalizations. Its contact and alarm call — the iconic "chick-a-dee-dee-dee" — varies in complexity based on context and threat level. Its territorial song is a clear, whistled two-note "fee-bee" (or "fee-bee-bee"), the first note higher than the second, heard most often on mild late-winter days as males begin to establish breeding territories. Chickadees also produce a variety of soft gargling calls within their flocks.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The Black-capped Chickadee remains abundant across its vast range, with an estimated population exceeding 100 million individuals. It adapts well to suburban and urban environments and benefits from nest box programs and winter feeding.

FEEDER TIPS

- Tube feeders or small hopper feeders work best
- Black-oil sunflower seeds are strongly preferred over striped
- Chickadees can be hand-tamed with patience over several weeks
- Place a nest box (1 1/8" hole diameter) near woodland edge
- Suet feeders are especially valued from October through March

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown & bib	Deep black
Cheek patches	Pure white
Back & wings	Slate gray
Belly	White to pale gray
Flanks	Buffy-white
Bill	Short, black

COLOR ME!

Black-capped Chickadee • *Poecile atricapillus* • Module 02



MODULE 03 / 40

Blue Jay

*Cyanocitta cristata***QUICK STATS**

Length	9.8 – 11.8 in (25 – 30 cm)
Wingspan	13.4 – 16.9 in (34 – 43 cm)
Weight	2.5 – 3.5 oz (70 – 100 g)
Lifespan	Up to 26 years (avg. 7 in wild)

PRESENTATION

The Blue Jay is one of the most striking and intelligent birds to visit North American gardens. Its bold blue, white, and black plumage makes it instantly recognizable at any feeder. Despite its reputation as a bully — it dominates most other species at feeding stations — the Blue Jay plays a vital ecological role as one of the continent's most important seed dispersers. Highly adaptable and extraordinarily vocal, it produces a wide variety of calls including convincing imitations of hawks, which it uses to clear feeders of competing birds.

HABITAT & RANGE

The Blue Jay is found throughout the eastern half of North America, from southern Canada (Alberta east to Nova Scotia and Newfoundland) south through Florida and west to the edge of the Great Plains. It inhabits a wide range of wooded environments including deciduous and mixed forests, forest edges, suburban neighborhoods, city parks, and gardens with mature oak trees. Northern populations are partially migratory, with large flocks moving south in autumn along coastlines and ridges, though many individuals remain year-round even in cold climates.

DIET & FEEDING

- Acorns — by far the most important food source
- Whole and shelled peanuts — a feeder favorite
- Sunflower seeds (black-oil and striped)
- Corn (cracked or whole kernel)
- Insects, grasshoppers, beetles during summer
- Occasionally: eggs and nestlings of other birds

BEHAVIOR & SOCIAL LIFE

Blue Jays are highly social outside the breeding season, gathering in loose flocks and traveling together through the forest canopy. They are among the most vocal birds in North America, producing a rich repertoire of calls including their signature loud "jay-jay-jay" alarm, musical liquid gurgles, whisper songs, and remarkably accurate mimicry of Red-shouldered and Red-tailed Hawks. This mimicry serves multiple purposes: alerting other jays to hawk presence and, controversially, potentially clearing feeders so the jay can feed undisturbed. Blue Jays have demonstrated problem-solving abilities and tool use in laboratory settings, ranking among the most cognitively advanced birds studied.

NESTING & BREEDING

Blue Jays form monogamous pairs that often mate for life. Both sexes participate in nest building, constructing a bulky open cup of twigs, bark strips, mud, and grass in the fork of a tree or shrub, typically 10–25 feet above ground. Clutches of 2–7 eggs (usually 4–5) are incubated for 17–18 days by both parents. Nestlings are fed by both parents and fledged at 17–21 days. Pairs typically raise one brood per year. Both parents aggressively mob and chase away predators, including owls, hawks, cats, and humans.

SONG & COMMUNICATION

The Blue Jay's vocal repertoire is one of the most varied of any North American bird. Its primary call is a loud, harsh "jay" or "jeer," used as an alarm. It also produces a bell-like "tool-tool" call, soft musical whisper songs audible only at close range, and uncannily accurate imitations of at least three hawk species. Researchers believe jays use hawk mimicry both as a warning system and as a competitive strategy at feeders.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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Blue Jay populations are stable across their range, with an estimated 45 million individuals in North America. The species has benefited from suburban expansion and the planting of oak trees in urban environments.

FEEDER TIPS

- Platform feeders or large hopper feeders accommodate their size
- Whole peanuts in the shell are irresistible and provide great viewing
- Scatter cracked corn on the ground — jays readily feed at ground level
- Plant oak trees: a single oak can produce thousands of acorns annually
- Expect jays to cache food — they may visit and depart rapidly with full beaks

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crest & upper wings	Bright blue with black bars
Tail	Blue with black barring, white tips
Face & nape	White with bold black necklace
Breast & belly	White to pale gray
Back	Blue-gray
Bill & legs	Black

COLOR ME!

Blue Jay • *Cyanocitta cristata* • Module 03



MODULE 04 / 40

House Finch

*Haemorhous mexicanus***QUICK STATS**

Length	5.0 – 6.0 in (12.7 – 15.2 cm)
Wingspan	8.0 – 10.0 in (20.3 – 25.4 cm)
Weight	0.6 – 0.9 oz (16 – 27 g)
Lifespan	Up to 11 years (avg. 2–3 in wild)

PRESENTATION

The House Finch is one of the most abundant and widespread songbirds in North America, found from coast to coast in virtually every habitat where humans have settled. Originally native to the western United States and Mexico, it was introduced to the eastern US in the 1940s when illegal pet trade birds were released on Long Island, New York. It has since colonized the entire continent with remarkable speed. The male's rosy-red head and breast make it one of the most colorful regular visitors to backyard feeders, while the streaked brown female is a master of camouflage.

HABITAT & RANGE

Today the House Finch is found across all 48 contiguous United States, southern Canada, and Mexico. It thrives in urban and suburban environments, farmlands, forest edges, and desert scrub. Highly adaptable, it nests on building ledges, in ornamental conifers, hanging baskets, and even inside cacti in the Southwest. Unlike many finches, it does not undertake long migrations, remaining year-round across most of its range.

DIET & FEEDING

- Nyjer (thistle) seed — strongly preferred
- Black-oil sunflower seeds and hulled chips
- White millet scattered on platform feeders
- Wild berries: mulberry, pyracantha, crabapple
- Flower buds and plant shoots in spring
- Weed seeds from dandelion, chickweed, and thistle

BEHAVIOR & SOCIAL LIFE

House Finches are highly gregarious birds that gather in flocks of dozens or even hundreds outside the breeding season. They are cheerful, active, and vocal at feeders, often tolerating close human approach. Males sing a long, rambling warble year-round — one of the few North American songbirds that sings regularly through winter. Fascinatingly, the intensity of the male's red coloration is entirely diet-dependent: birds consuming more carotenoid-rich foods (berries, certain plants) develop brighter red plumage, while poorly-fed males may appear orange or yellow. Females actively choose the reddest males as mates, using color as a reliable indicator of foraging ability and health.

NESTING & BREEDING

House Finches are remarkably flexible nesters, using an extraordinary range of sites including tree cavities, cliff ledges, building eaves, hanging flower baskets, and old nests of other species. The female builds a compact cup of grasses, leaves, and fine plant fibers. Clutches of 2–6 pale blue eggs with dark speckles are incubated for 13–14 days by the female alone. The male feeds her during incubation. Nestlings fledge at 12–19 days. Pairs raise 2–3 broods per year, sometimes beginning a new nest before the previous brood has fully fledged.

SONG & COMMUNICATION

The male House Finch produces one of the most melodious and sustained songs of any common feeder bird — a rich, cheerful warble lasting 3 seconds or more, delivered from a prominent perch. Song is heard year-round but peaks in spring. The call is a sharp "cheep" often given in flight. Flocks maintain contact with soft chirping notes while foraging together.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With an estimated population of over 267 million individuals, the House Finch is one of the most numerous birds in North America. However, populations in the eastern US suffered dramatic declines in the 1990s due to a contagious eye disease (Mycoplasmal conjunctivitis) that spread rapidly through feeder populations. Numbers have since partially recovered. If you see a finch with red, swollen, or crusty eyes at your feeder, clean feeders immediately with a 10% bleach solution.

FEEDER TIPS

- Tube feeders with small ports are ideal for finches
- Nyjer seed in a dedicated thistle sock attracts large flocks
- Clean feeders regularly — House Finches are susceptible to eye disease
- Platform feeders with millet attract ground-feeding individuals
- House Finches often feed in large, noisy groups — expect crowds

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head & breast (male)	Raspberry red to orange-red
Back & wings	Brown with dark streaking
Belly	White with brown streaks
Rump (male)	Reddish-pink
Female (entire)	Plain brown, heavily streaked
Bill	Short, conical, gray-brown

COLOR ME!

House Finch • *Haemorhous mexicanus* • Module 04



MODULE 05 / 40

Ruby-throated Hummingbird

*Archilochus colubris***QUICK STATS**

Length	2.8 – 3.5 in (7 – 9 cm)
Wingspan	3.1 – 4.3 in (8 – 11 cm)
Weight	0.07 – 0.21 oz (2 – 6 g)
Lifespan	Up to 9 years (avg. 3–5 in wild)

PRESENTATION

The Ruby-throated Hummingbird is the only hummingbird species that regularly breeds in eastern North America, and one of the most extraordinary birds on the continent. Weighing less than a nickel, it is a marvel of biological engineering: its wings beat 53 times per second, its heart beats up to 1,200 times per minute during flight, and it is the only bird capable of sustained backwards flight. The male's iridescent ruby-red gorget (throat patch) can appear jet black in poor lighting, then flash brilliant red as the angle of light changes — one of nature's most spectacular optical effects.

HABITAT & RANGE

During summer, the Ruby-throated Hummingbird breeds across the entire eastern half of North America, from Nova Scotia and Manitoba south through Florida and west to the Great Plains. It inhabits forest edges, gardens, orchards, and meadows with abundant flowering plants. Each fall, it undertakes one of the most astonishing migrations in the bird world: a non-stop flight of 500–900 miles across the Gulf of Mexico to its wintering grounds in Mexico and Central America. It returns north each spring from April through May.

DIET & FEEDING

- Flower nectar — trumpetvine, salvia, monarda, impatiens
- Feeder solution: 1 part white sugar to 4 parts water (no dye)
- Tiny insects and spiders — essential protein source
- Tree sap from sapsucker wells (used opportunistically)
- Never use honey, artificial sweeteners, or red dye in feeders

BEHAVIOR & SOCIAL LIFE

Ruby-throated Hummingbirds are among the most aggressively territorial birds in North America relative to their size. Males fiercely defend feeding territories, dive-bombing intruders — including birds many times their size — with surprising ferocity. Outside of courtship and mating, males and females live entirely solitary lives. To survive cold nights and food shortages, hummingbirds enter a state of torpor: their body temperature drops from 40°C to around 18°C, their heart rate falls from 1,200 to just 50 beats per minute, and their breathing nearly stops. This metabolic shutdown can save up to 60% of their nightly energy expenditure.

NESTING & BREEDING

The female alone builds one of nature's most remarkable nests: a walnut-sized cup of plant down and spider silk, camouflaged on the outside with lichens, and expandable as the chicks grow. It is typically placed on a slender downward-sloping branch 10–20 feet above ground. She lays exactly 2 white eggs the size of peas, incubating them alone for 11–14 days. She feeds the nestlings by inserting her bill deep into their throats and regurgitating nectar and insects. Young fledge at 18–22 days. The male plays no role in nesting or chick-rearing.

SONG & COMMUNICATION

Ruby-throated Hummingbirds are not known for melodious song. Their primary vocalizations are sharp, rapid chips and chattering calls used during territorial disputes and pursuit chases. Males perform spectacular U-shaped or pendulum dive displays during courtship, producing a distinctive wing-buzz at the bottom of each arc. The wingbeat itself produces the characteristic humming sound that gives the family its name.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With a stable population estimated at nearly 34 million individuals, the Ruby-throated Hummingbird is not currently threatened. However, it faces pressure from habitat loss on its wintering grounds in Central America and from climate change affecting the timing of flower blooms relative to migration. Planting native nectar plants and maintaining clean feeders are the most impactful ways to support local populations.

FEEDER TIPS

- Use a red feeder — no need to add dye to the solution
- Recipe: boil 1 cup white sugar in 4 cups water, cool completely
- Change solution every 2–3 days in hot weather to prevent fermentation
- Clean feeders with hot water and a bottle brush — no soap needed
- Hang feeders by late April in the South, mid-May in the North
- Plant trumpet vine, bee balm, salvia, and cardinal flower nearby

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Back & crown (both)	Iridescent emerald green
Gorget — male throat	Ruby red (may look black)
Breast & belly	White to pale gray
Tail (male)	Forked, dark green-black
Female throat	White, no red gorget
Bill	Long, straight, black

COLOR ME!

Ruby-throated Hummingbird • *Archilochus colubris* • Module 05



MODULE 06 / 40

American Robin

*Turdus migratorius***QUICK STATS**

Length	7.9 – 11.0 in (20 – 28 cm)
Wingspan	12.2 – 15.7 in (31 – 40 cm)
Weight	2.7 – 3.0 oz (77 – 85 g)
Lifespan	Up to 14 years (avg. 2 in wild)

PRESENTATION

The American Robin is one of the most familiar and beloved birds in North America — a true symbol of spring whose return each year is eagerly anticipated across the continent. Plump, upright, and boldly patterned with its signature orange-red breast, it is a consummate garden bird, foraging confidently across lawns for earthworms and nesting on ledges, trees, and even window sills in close proximity to humans. Despite being called a robin, it is actually a thrush — a member of the same family as bluebirds and solitaires — and is the largest thrush in North America.

HABITAT & RANGE

The American Robin breeds across virtually all of North America, from Alaska and northern Canada south through Mexico, making it one of the continent's most widespread birds. It inhabits an extraordinary range of environments: lawns, parks, forests, mountains up to the treeline, tundra edges, and urban centers. Northern populations migrate south in autumn, though many winter in the southern US where berry-laden trees provide food. Year-round residents are common throughout much of the contiguous United States. Large winter flocks of thousands can gather in roosts.

DIET & FEEDING

- Earthworms — located by sight and sound, not smell
- Wild berries: holly, crabapple, juniper, dogwood, cedar
- Insects, beetles, grasshoppers, and caterpillars
- Fruit: grapes, cherries, plums, serviceberry
- Rarely visits seed feeders — prefers open ground foraging
- Plant berry-producing shrubs to attract robins to your garden

BEHAVIOR & SOCIAL LIFE

American Robins are highly visible and active birds, spending much of their time foraging on the ground with a characteristic run-stop-tilt posture as they listen and watch for earthworms. Contrary to popular belief, they locate worms primarily by sight rather than hearing, though both senses play a role. Outside breeding season, robins are highly gregarious, forming communal roosts of thousands or even millions of birds in woodlands. These roosts disperse each morning as birds fan out to forage, then reconvene at dusk. Males arrive on breeding grounds before females each spring and immediately begin singing to establish territories.

NESTING & BREEDING

The American Robin is one of the earliest nesters in North America, often beginning as soon as late February in the South. The female builds a robust cup nest of grasses, twigs, and worm castings, reinforced with mud and lined with fine grasses. Nests are placed on building ledges, in tree forks, or on fence posts 5–15 feet above ground. The female lays 3–5 eggs of a vivid, unmistakable sky blue — the origin of the color name "robin's egg blue." Incubation lasts 12–14 days. Both parents feed nestlings, which fledge at 13 days. Pairs typically raise 2–3 broods per season.

SONG & COMMUNICATION

The robin's song is one of the most recognized sounds of the North American spring — a rich, melodious caroling of rising and falling phrases often transcribed as "cheerily, cheer-up, cheerio." Males begin singing before dawn and continue into dusk, sometimes even singing after dark near artificial lights. The sharp "tut-tut-tut" alarm call warns of ground predators, while a high thin "see" alerts to aerial threats. Robins are among the first birds to sing in the morning chorus.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With an estimated population of 300 million individuals, the American Robin is one of the most abundant birds in North America. Its adaptability to human-modified landscapes has helped it thrive despite widespread habitat changes. It does face localized threats from pesticide use, which reduces earthworm availability and can cause direct poisoning.

FEEDER TIPS

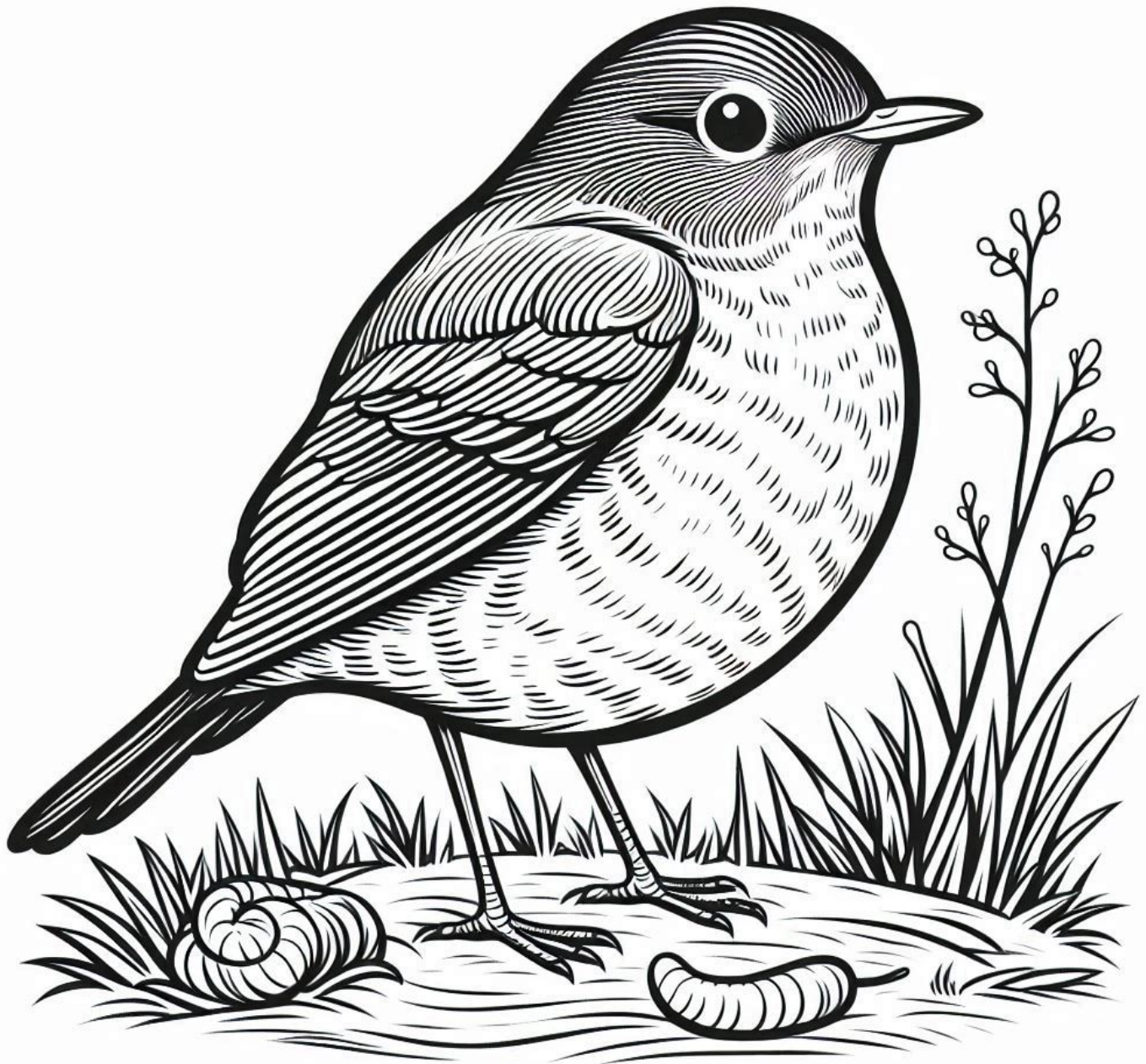
- Robins rarely use seed feeders — attract them differently
- Install a shallow birdbath: robins love to bathe and drink
- Plant holly, crabapple, dogwood, or serviceberry for winter berries
- Avoid pesticides on lawns — robins depend on healthy earthworm populations
- A platform feeder with mealworms or fruit pieces may attract them

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head & back	Dark gray to black
Breast & belly	Brick orange-red
Throat	White with dark streaks
White eye-ring	Pure white crescent
Bill	Yellow-orange
Legs & feet	Dark brown-gray

COLOR ME!

American Robin • *Turdus migratorius* • Module 06



MODULE 07 / 40

Downy Woodpecker

Dryobates pubescens

QUICK STATS

Length	5.5 – 6.7 in (14 – 17 cm)
Wingspan	9.8 – 11.8 in (25 – 30 cm)
Weight	0.7 – 1.0 oz (20 – 33 g)
Lifespan	Up to 11 years (avg. 1–2 in wild)

PRESENTATION

The Downy Woodpecker is the smallest woodpecker in North America and one of the most familiar visitors to backyard feeders across the continent. Despite its diminutive size, it is a bold and active bird, clinging to suet cages, sunflower feeders, and tree trunks with equal ease. Its crisp black-and-white plumage and the male's bright red nape patch make it easy to identify. The Downy is often confused with the larger Hairy Woodpecker, but can be distinguished by its much shorter bill relative to head size and its smaller overall dimensions. It is one of the most widespread woodpeckers in North America and a year-round resident throughout its range.

HABITAT & RANGE

The Downy Woodpecker inhabits a remarkable range of forested and semi-open environments across North America, from Alaska and northern Canada south through most of the United States, absent only from the arid Southwest and parts of the Great Plains. It thrives in deciduous and mixed forests, orchards, riparian corridors, suburban parks, and gardens with mature trees. Unlike many woodpeckers, it readily adapts to fragmented and urban habitats and is a common year-round feeder visitor across most of its range.

DIET & FEEDING

- Suet cakes — the single most effective food to offer
- Black-oil sunflower seeds and hulled chips
- Peanut butter (plain, no additives) smeared on bark
- Bark insects: beetle larvae, ants, moth cocoons
- Wild berries and plant galls in winter
- Poison ivy berries — an important winter food source

BEHAVIOR & SOCIAL LIFE

Downy Woodpeckers are active, acrobatic birds that forage across a wide variety of surfaces — tree trunks, large branches, weed stems, and even corn stalks. Their small size allows them to exploit foraging niches unavailable to larger woodpeckers, including the tips of small branches and slender plant stems where insect eggs and larvae overwinter. Outside breeding season, Downies frequently join mixed-species foraging flocks with chickadees, nuthatches, and kinglets, benefiting from the collective vigilance of the group. Males and females partition foraging habitat even within a territory: males tend to forage on smaller branches and weed stems, females on larger trunks.

NESTING & BREEDING

Downy Woodpeckers excavate their own nest cavities in dead or dying trees, a process that takes both sexes 1–3 weeks to complete. The entrance hole is perfectly round, about 1.25 inches in diameter, leading to a gourd-shaped chamber 6–12 inches deep lined only with wood chips. No nest material is added. Clutches of 3–8 white eggs are incubated by both parents for 12 days. The male incubates at night. Nestlings are fed by both parents and fledge at 20–25 days. Old nest cavities are vital for other cavity-nesting species including chickadees, bluebirds, and small owls.

SONG & COMMUNICATION

The Downy Woodpecker's primary call is a sharp, flat "pik" — shorter and higher-pitched than the similar Hairy Woodpecker's call. It also produces a descending whinny call used during territorial interactions and a rattling "drum" — a rapid series of bill strikes on a resonant dead branch — used to establish territory and attract mates. Drumming rates average about 17 strikes per second. Both sexes drum, though males drum more frequently.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The Downy Woodpecker population is estimated at over 14 million individuals and remains stable across its range. It benefits from dead tree retention in managed forests and suburban areas. Its excavated nest cavities provide essential housing for dozens of other cavity-dependent species, making it a keystone species in many woodland ecosystems.

FEEDER TIPS

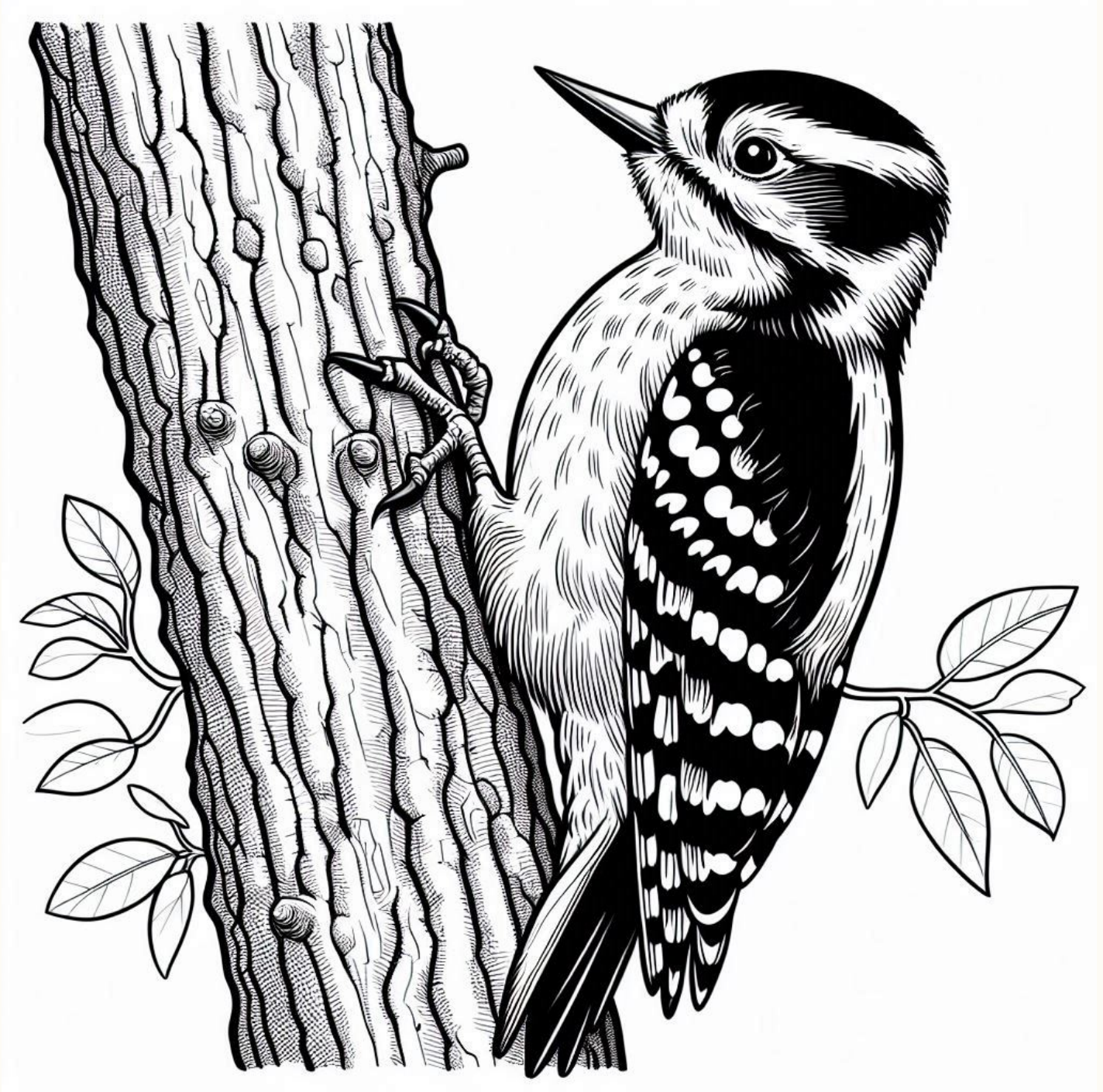
- Suet cages are the most reliable way to attract Downy Woodpeckers
- Offer suet year-round — especially critical in cold winters
- Upside-down suet feeders deter starlings while Downies feed easily
- Hang feeders near tree trunks — Downies feel safer close to cover
- Leave dead trees standing when safe — essential for nesting and foraging

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown & back	<i>Solid black</i>
Wing spots	<i>Black with white dots/bars</i>
White back stripe	<i>Pure white center stripe</i>
Underparts	<i>White to pale buff</i>
Red nape patch (male)	<i>Bright red — rear of crown only</i>
Bill	<i>Short, dark gray, chisel-shaped</i>

COLOR ME!

Downy Woodpecker • *Dryobates pubescens* • Module 07



MODULE 08 / 40

White-breasted Nuthatch

*Sitta carolinensis***QUICK STATS**

Length	5.1 – 5.5 in (13 – 14 cm)
Wingspan	7.9 – 10.6 in (20 – 27 cm)
Weight	0.6 – 1.1 oz (18 – 30 g)
Lifespan	Up to 10 years (avg. 2 in wild)

PRESENTATION

The White-breasted Nuthatch is one of the most distinctive and entertaining birds to visit backyard feeders — instantly recognizable by its habit of walking headfirst down tree trunks, a behavior unique among North American birds. This remarkable feat is made possible by an unusually large hind toe that provides a powerful grip regardless of orientation. Compact, neckless, and boldly patterned in black, white, and blue-gray, it is a year-round resident that pairs for life and maintains a fixed territory through all seasons. Its loud, nasal calls are among the most recognizable winter sounds in eastern North American forests.

HABITAT & RANGE

The White-breasted Nuthatch is found across most of the United States and southern Canada, from British Columbia east to Nova Scotia and south through the Appalachians and into Mexico. It inhabits mature deciduous and mixed forests, forest edges, orchards, parks, and wooded suburban gardens with large trees. It shows a strong preference for oak, hickory, and beech-dominated forests where its primary wild food — acorns and nuts — is abundant. It is a permanent resident throughout its range and does not migrate.

DIET & FEEDING

- Sunflower seeds — black-oil and striped both accepted
- Suet cakes — especially valued in winter
- Whole and shelled peanuts
- Acorns and hickory nuts (cached in bark crevices)
- Insects and spiders gleaned from bark year-round
- Peanut butter mixtures smeared into bark furrows

BEHAVIOR & SOCIAL LIFE

White-breasted Nuthatches are monogamous and maintain year-round pair bonds, with mates staying in close contact through constant soft calls. They are vigorously territorial, with pairs defending the same woodland territory across multiple years. Their signature head-down foraging posture gives them a unique advantage: they can spot insects and cached food items hidden in bark crevices from angles that upward-foraging birds like woodpeckers would miss entirely. They are avid food catchers, wedging seeds and nuts into bark furrows and covering them with lichen or bark fragments — a behavior that has earned them the folk name "Devil Downhead" in some regions. They often join winter foraging flocks with chickadees and Downy Woodpeckers.

NESTING & BREEDING

Breeding begins in March or April. Pairs use natural tree cavities, old woodpecker holes, or nest boxes, with the female doing most of the nest building. She constructs a cup of bark strips, grasses, and fur inside the cavity. A fascinating defensive behavior: the pair rubs crushed insects (particularly blister beetles) around the entrance hole — the noxious chemicals may deter squirrels and other predators. Clutches of 5–9 white eggs speckled with red-brown are incubated by the female for 13–14 days. Both parents feed the nestlings, which fledge at 26 days — an unusually long nestling period.

SONG & COMMUNICATION

The White-breasted Nuthatch is one of the most vocal winter birds. Its primary call is a loud, nasal "yank yank yank" — a rapid series that carries well through winter woodland. A softer, higher "ip" contact call keeps pairs in touch while foraging. The male's song, heard mostly in late winter and early spring, is a rapid series of low whistled notes "whi-whi-whi-whi" repeated several times. Females also vocalize frequently with soft nasal calls.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The White-breasted Nuthatch population is estimated at approximately 10 million individuals and is considered stable. It benefits from the retention of large mature trees with natural cavities, and from nest box programs in managed forests. Forest fragmentation and the loss of old-growth deciduous woodland are the primary long-term concerns for this species.

FEEDER TIPS

- Offer sunflower seeds in a tube or hopper feeder near large trees
- A suet cage mounted on a tree trunk is ideal for nuthatches
- Smear peanut butter into bark crevices — they will find it quickly
- Install a nest box with a 1.25–1.5 inch entrance hole on a large tree
- Nuthatches cache seeds — place feeders near rough-barked oaks or maples

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown & nape (male)	Glossy black cap
Crown (female)	Dark gray — not black
Back & wings	Blue-gray
Face & underparts	White to pale
Flanks & undertail	Rusty chestnut-orange
Bill	Long, pointed, dark gray

COLOR ME!

White-breasted Nuthatch • *Sitta carolinensis* • Module 08



MODULE 09 / 40

American Goldfinch

*Spinus tristis***QUICK STATS**

Length	4.3 – 5.1 in (11 – 13 cm)
Wingspan	7.5 – 8.7 in (19 – 22 cm)
Weight	0.4 – 0.7 oz (11 – 20 g)
Lifespan	Up to 11 years (avg. 3–6 in wild)

PRESENTATION

The American Goldfinch is one of the most brilliantly colored birds in North America — the male's summer plumage of blazing canary yellow and jet black is unmistakable at any feeder. Known affectionately as the "wild canary," it is one of the few finches that undergoes a complete molt twice a year, transitioning from dazzling yellow in summer to subdued olive-brown in winter. It is also one of the latest nesters in North America, deliberately timing its breeding to coincide with the peak availability of thistle seeds in late summer — the primary food used to feed its nestlings.

HABITAT & RANGE

The American Goldfinch breeds across southern Canada and the northern two-thirds of the United States, from British Columbia east to Newfoundland and south through the Appalachians. In winter, northern populations migrate south, and the species can then be found across the entire continental US and into northern Mexico. It inhabits open and semi-open environments: weedy fields, meadows, floodplains, roadsides, forest edges, orchards, and suburban gardens. It shows a strong preference for areas with abundant thistles, sunflowers, and other composite flowers.

DIET & FEEDING

- Nyjer (thistle) seed — the top choice, attracts large flocks
- Black-oil sunflower seeds and hulled chips
- Native thistle, coneflower, and sunflower seeds in the wild
- Dandelion, aster, and goldenrod seeds
- Almost entirely vegetarian — rarely eats insects
- Birch and alder catkins in late winter

BEHAVIOR & SOCIAL LIFE

American Goldfinches are highly social birds that travel in flocks year-round, even during the breeding season when most songbirds become territorial. Flocks can number in the hundreds at productive feeding sites. Their flight pattern is distinctively undulating — a series of flaps followed by a glide — and they frequently call in flight with a cheerful "per-chick-o-ree" that announces their presence overhead. Males are loosely territorial around nest sites but tolerate other goldfinches feeding nearby. They are among the most acrobatic feeder birds, clinging upside-down to thistle socks and swinging seed heads with ease.

NESTING & BREEDING

The American Goldfinch is the latest nesting songbird in North America, typically not beginning until late June or July — months after most other species have already fledged young. This timing is not accidental: it synchronizes perfectly with the peak ripening of thistle and other composite seeds. The female weaves an exceptionally tight, compact cup nest of plant fibers and thistle down, bound with spider silk, in the fork of a shrub or small tree. The nest is so tightly woven it can hold water. She lays 4–6 pale blue eggs, incubating them alone for 12–14 days while the male feeds her. Both parents feed nestlings regurgitated seeds. Young fledge at 11–17 days.

SONG & COMMUNICATION

The goldfinch's song is a long, rambling series of musical twitters, warbles, and trills — bright and cheerful, delivered from a perch or in flight. The flight call "per-chick-o-ree" or "po-ta-to-chip" is one of the most recognizable sounds of open country in summer. A soft "swee" contact call keeps flock members in touch. Males sing most actively from late spring through summer.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining slightly
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With a population estimated at 42 million individuals, the American Goldfinch remains common but has shown gradual declines in recent decades, linked to loss of weedy meadow habitat, agricultural intensification, and the reduction of native thistles. Planting native wildflowers — especially coneflowers, black-eyed Susans, and native thistles — is one of the most effective ways to support goldfinch populations.

FEEDER TIPS

- Nyjer (thistle) seed in a dedicated thistle sock or tube feeder
- Expect large flocks — install multiple feeders to reduce competition
- Keep nyjer fresh — it goes stale quickly and goldfinches will reject it
- Plant native coneflowers and leave seed heads standing through winter
- Goldfinches arrive later in the day — most active mid-morning to afternoon

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Body (male, summer)	Brilliant canary yellow
Wings & tail	Black with white wing bars
Forehead patch (male)	Black cap — front only
Body (female)	Olive-yellow, duller than male
Body (male, winter)	Olive-brown — loses bright yellow
Bill	Short, conical, orange-pink

COLOR ME!

American Goldfinch • *Spinus tristis* • Module 09



MODULE 10 / 40

Dark-eyed Junco

Junco hyemalis

QUICK STATS

Length	5.5 – 6.3 in (14 – 16 cm)
Wingspan	7.1 – 9.8 in (18 – 25 cm)
Weight	0.6 – 1.1 oz (18 – 30 g)
Lifespan	Up to 11 years (avg. 3 in wild)

PRESENTATION

The Dark-eyed Junco is one of the most abundant birds in North America and a quintessential winter feeder visitor, earning its nickname "snowbird" for its tendency to appear at feeders just as the first cold weather arrives. Crisp and tidy in its dark gray-and-white plumage, it is among the easiest birds to identify — the sudden flash of white outer tail feathers as it flies away is one of the most recognizable field marks in North American birding. The Dark-eyed Junco is actually a complex of several regional forms that were once considered separate species, including the Slate-colored, Oregon, Pink-sided, and White-winged Juncos.

HABITAT & RANGE

The Dark-eyed Junco breeds across the boreal forests of Canada and Alaska, and in montane coniferous forests south through the Appalachians, Rockies, and Sierra Nevada. In winter it migrates south and can be found across virtually the entire contiguous United States, making it one of the most widespread winter birds on the continent. It favors woodland edges, brushy areas, and open ground near cover. In the breeding season it inhabits mature coniferous and mixed forests, often at higher elevations.

DIET & FEEDING

- White millet — strongly preferred, especially on the ground
- Black-oil sunflower seeds and cracked corn
- Milo, sorghum, and mixed seed blends
- Weed seeds: chickweed, sorrel, buckwheat
- Insects and spiders during breeding season
- Juncos are primarily ground feeders — scatter seed on the ground

BEHAVIOR & SOCIAL LIFE

Dark-eyed Juncos are highly social in winter, forming flocks of 10–30 birds that maintain a clear dominance hierarchy. Males dominate females, and adults dominate young birds. Within a flock, subordinate individuals are pushed to the edges of feeding areas where predator risk is higher. Despite this hierarchy, flock members cooperate in vigilance — when one bird detects a threat and flushes, the flash of white tail feathers from multiple birds simultaneously serves as a visual alarm signal. Juncos are primarily ground feeders, hopping and scratching through leaf litter and snow with both feet simultaneously in a characteristic double-scratch motion.

NESTING & BREEDING

Juncos nest on or near the ground, often tucking their cup nest under an overhanging root, rock ledge, or clump of vegetation on a steep slope or bank. The female builds a neat cup of grasses, moss, and bark strips, lined with fine grasses and hair. Clutches of 3–5 eggs (pale bluish or grayish, spotted with brown) are incubated by the female for 12–13 days. Both parents feed the nestlings, which fledge at just 9–13 days — unusually quickly, likely an adaptation to reduce nest predation risk. Pairs raise 2–3 broods per season.

SONG & COMMUNICATION

The junco's song is a simple, pleasant musical trill on one pitch, similar to a Chipping Sparrow but slower and more musical. It is heard primarily on the breeding grounds. In winter flocks, juncos communicate with a variety of sharp "tick" and "smacking" call notes, as well as soft twittering calls within the flock. The sharp alarm call triggers the characteristic tail-flashing response.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With an estimated population of 630 million individuals, the Dark-eyed Junco is one of the most numerous birds in North America. It is estimated that juncos make up roughly 71% of all birds at feeders across the US during winter. Population trends are stable across most of its range.

FEEDER TIPS

- Scatter white millet directly on the ground — juncos prefer ground feeding
- A low platform feeder or ground tray works better than tube feeders
- Clear snow from feeding areas — juncos struggle to dig through deep snow
- Dense shrubs nearby provide essential escape cover from hawks
- Juncos often arrive in mixed flocks with sparrows and chickadees

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Hood & back (male)	Slate gray to dark charcoal
Belly	White — sharp contrast with hood
Outer tail feathers	White — flash when flying
Female hood	Paler brown-gray, less contrasted
Bill	Short, pale pink — conical
Legs	Pale pink

COLOR ME!

Dark-eyed Junco • *Junco hyemalis* • Module 10



MODULE 11 / 40

Song Sparrow

*Melospiza melodia***QUICK STATS**

Length	4.7 – 6.7 in (12 – 17 cm)
Wingspan	7.1 – 9.4 in (18 – 24 cm)
Weight	0.4 – 1.9 oz (11 – 53 g)
Lifespan	Up to 11 years (avg. 3 in wild)

PRESENTATION

The Song Sparrow is one of the most widespread and variable songbirds in North America, found in nearly every habitat from Alaska to Mexico. Despite its streaky brown plumage that many birders initially dismiss as "just a sparrow," it is a remarkably accomplished singer — the male begins his rich, complex song as early as February, often singing from an exposed perch even in snow and freezing temperatures. The Song Sparrow is also one of the most scientifically studied birds in the world, with research spanning over a century that has illuminated fundamental principles of ecology, behavior, and evolution.

HABITAT & RANGE

The Song Sparrow occupies an extraordinary range of habitats across North America, from the Aleutian Islands of Alaska south through Canada and across the entire continental US to central Mexico. It inhabits wetland edges, marshes, shrubby fields, forest edges, riparian thickets, coastal dunes, gardens, and suburban parks. It shows remarkable geographic variation: over 50 subspecies have been described, ranging from tiny pale desert birds to large dark subspecies in the Pacific Northwest. Many northern populations migrate south in winter, while southern and coastal populations are permanent residents.

DIET & FEEDING

- White millet — the most attractive seed at feeders
- Black-oil sunflower seeds and cracked corn
- Weed seeds: crabgrass, ragweed, smartweed, knotweed
- Insects and spiders — critical during breeding season
- Wild berries and small fruits in autumn
- Primarily forages on the ground or in low vegetation

BEHAVIOR & SOCIAL LIFE

Song Sparrows are loosely territorial and moderately social. Males defend breeding territories through song, often countersinging with neighboring males in elaborate vocal exchanges. Outside the breeding season they become more gregarious, joining mixed sparrow flocks. They forage primarily on the ground, scratching through leaf litter with a characteristic double-scratch hop. When startled, they fly low and briefly pump their rounded tail — a distinctive behavior that helps identify them in flight. Song Sparrows show strong site fidelity, returning to the same territories year after year.

NESTING & BREEDING

Song Sparrows are among the earliest breeders in North America, with males singing on territory as early as late January in the South. The female builds a well-hidden cup nest of grasses, weeds, and bark strips, placed on the ground or low in dense vegetation. Clutches of 3–5 pale greenish eggs with brown speckles are incubated by the female for 12–14 days. Both parents feed nestlings, which fledge at 10–12 days. Song Sparrows are frequent hosts of Brown-headed Cowbird eggs — a brood parasite that lays its eggs in other birds' nests. Pairs raise 2–3 broods per season.

SONG & COMMUNICATION

The Song Sparrow earns its name through one of the most complex and beautiful songs of any North American sparrow. Each male learns a repertoire of 8–20 distinct song types and switches between them throughout the day. Songs typically begin with 2–3 short notes, followed by a trill and then a complex jumble of notes. The song is often described as beginning with "Maids! Maids! Maids! Pick up your teakettle-ettle-ettle." Each male's songs are slightly different, allowing individual recognition by neighbors.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The Song Sparrow population is estimated at 130 million individuals and remains broadly stable, though some regional populations have declined due to wetland loss and habitat fragmentation. It is one of the most studied birds in the world — research on Song Sparrow populations on Mandarte Island, British Columbia has been ongoing since 1975.

FEEDER TIPS

- Scatter white millet on the ground or a low platform feeder
- Dense shrubs and brush piles near feeders provide essential cover
- Song Sparrows often arrive at feeders early in the morning
- A shallow ground-level birdbath is strongly attractive to them
- Leave native grasses and weedy patches in the garden for natural foraging

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown	Brown with gray central stripe
Back & wings	Brown, heavily streaked dark
Breast & sides	White with bold brown streaks
Central breast spot	Dark brown — diagnostic dot
Supercilium (eyebrow)	Gray-white stripe above eye
Bill	Short, conical, pale gray

COLOR ME!

Song Sparrow • *Melospiza melodia* • Module 11



MODULE 12 / 40

White-throated Sparrow

Zonotrichia albicollis

QUICK STATS

Length	6.3 – 7.1 in (16 – 18 cm)
Wingspan	7.9 – 9.1 in (20 – 23 cm)
Weight	0.8 – 1.1 oz (22 – 32 g)
Lifespan	Up to 9 years (avg. 3 in wild)

PRESENTATION

The White-throated Sparrow is a large, handsome sparrow and a beloved winter visitor to feeders across eastern and southern North America. Its bold head pattern — alternating black and white crown stripes, a tiny yellow spot between bill and eye, and a sharply defined white throat — makes it one of the easiest sparrows to identify. Its haunting, whistled song, often transcribed as "Oh sweet Canada Canada Canada," is one of the most evocative sounds of the northern forest. The species comes in two distinct color morphs — white-striped and tan-striped — that interbreed freely but almost always choose mates of the opposite morph.

HABITAT & RANGE

The White-throated Sparrow breeds in the boreal forests of Canada and the northeastern United States, from British Columbia east to Newfoundland, south through the northern Great Lakes states and New England. It winters across the eastern half of the United States south to Florida and the Gulf Coast, and in smaller numbers west to California. On its breeding grounds it inhabits the edges of boreal forests, regenerating clear-cuts, and shrubby bogs. In winter it frequents woodland edges, brushy thickets, hedgerows, and suburban gardens with dense shrub cover.

DIET & FEEDING

- White millet — the top choice at feeders
- Black-oil sunflower seeds and hulled chips
- Cracked corn scattered on the ground
- Wild weed seeds: ragweed, smartweed, knotweed
- Wild berries: dogwood, sumac, bayberry
- Insects and spiders during the breeding season

BEHAVIOR & SOCIAL LIFE

White-throated Sparrows are gregarious in winter, foraging in flocks of 10–50 birds, often mixed with Dark-eyed Juncos, Song Sparrows, and other ground-feeding species. They are primarily ground feeders, scratching vigorously through leaf litter with a characteristic double-foot kick to expose buried seeds. The two color morphs — white-striped and tan-striped — represent a remarkable genetic polymorphism. White-striped birds are more aggressive and sing more frequently; tan-striped birds are better parents. Remarkably, nearly all mated pairs consist of one bird of each morph, a pattern maintained across the entire species.

NESTING & BREEDING

White-throated Sparrows nest on or near the ground in dense boreal shrub habitat. The female builds a cup nest of grasses, moss, and pine needles, hidden beneath shrubs or at the base of small trees. Clutches of 4–5 pale blue or greenish eggs with brown spotting are incubated by the female for 11–14 days. Both parents feed the nestlings, which fledge at 7–12 days. The species raises 1–2 broods per season. It is a frequent cowbird host on its breeding grounds.

SONG & COMMUNICATION

The White-throated Sparrow's song is one of the most distinctive and beloved in North American birding — a clear, pure whistle often rendered as "Old Sam Peabody Peabody Peabody" or "Oh sweet Canada Canada Canada." It consists of 2–3 long introductory notes followed by a series of triplets on a different pitch. Males sometimes sing on warm winter days, though song peaks on the breeding grounds. The call is a sharp, metallic "chink."

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining
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The White-throated Sparrow population is estimated at 140 million individuals but has declined significantly — by roughly 30% — since 1970, largely due to loss of boreal forest breeding habitat from logging and climate change. Maintaining dense shrub cover in winter gardens and reducing window collisions are meaningful ways to help this species.

FEEDER TIPS

- Scatter white millet on the ground beneath feeders
- Dense shrub thickets nearby are essential — they retreat to cover constantly
- Low platform feeders or ground trays are preferred over tube feeders
- White-throated Sparrows often linger at feeders well into spring
- Reduce window collisions with decals — this species is particularly vulnerable

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown stripes	Bold black & white alternating
Yellow lore spot	Bright yellow — above bill/eye
Throat patch	Clean white — sharply defined
Back & wings	Brown, streaked dark
Breast & belly	Gray-white, unstreaked
Bill	Short, conical, pale gray-pink

COLOR ME!

White-throated Sparrow • *Zonotrichia albicollis* • Module 12



MODULE 13 / 40

Mourning Dove

*Zenaida macroura***QUICK STATS**

Length	9.1 – 13.4 in (23 – 34 cm)
Wingspan	17.7 in (45 cm)
Weight	3.4 – 6.0 oz (96 – 170 g)
Lifespan	Up to 31 years (avg. 1–2 in wild)

PRESENTATION

The Mourning Dove is one of the most abundant and widespread birds in North America, with a population estimated at 350 million individuals — making it the most numerous wild bird on the continent. Elegant in its simplicity, its soft buffy-tan plumage, long pointed tail, and gentle demeanor make it an instant favorite at feeders. Its mournful, cooing call — "ooo-woo-woo-woo" — is one of the most evocative and widely recognized sounds of the North American countryside, often mistaken for an owl by those who first hear it. It is also one of the most important game birds in North America, legally hunted in most states.

HABITAT & RANGE

The Mourning Dove is found year-round across the entire contiguous United States, southern Canada, Mexico, and the Caribbean. It is a bird of open and semi-open habitats: farmland, grasslands, woodland edges, roadsides, suburban parks, and gardens. It avoids dense, closed-canopy forest but thrives wherever there are open ground areas for foraging and scattered trees or shrubs for nesting and roosting. Northern populations are partially migratory, moving south in cold winters, while southern birds are permanent residents.

DIET & FEEDING

- White millet — the top preference, eagerly consumed on the ground
- Black-oil sunflower seeds and cracked corn
- Safflower seeds and milo
- Wild grass seeds, weed seeds, and waste grain
- Occasionally pine seeds and small berries
- Grit (small stones) swallowed to aid digestion in the crop

BEHAVIOR & SOCIAL LIFE

Mourning Doves are gregarious birds that gather in large flocks at productive feeding sites, sometimes numbering in the dozens or even hundreds. They are ground feeders almost exclusively, walking with a characteristic bobbing head motion as they pick up seeds. Unlike most songbirds, they swallow seeds whole and store them in an enlarged crop, then digest them later — allowing them to feed rapidly and retreat to cover. Their wings produce a loud, distinctive whistling sound when they take off — a built-in alarm signal for the flock. Despite their gentle appearance, males can be aggressively competitive at feeding areas, chasing rivals with bowing displays.

NESTING & BREEDING

Mourning Doves are prolific breeders, raising up to 6 broods per year in warm climates — more than almost any other North American bird. The nest is a remarkably flimsy platform of twigs, so loosely constructed that eggs are often visible from below. It is placed in a tree, shrub, or even on the ground. Both parents incubate 2 pure white eggs for 13–15 days. Both parents feed nestlings "crop milk" — a protein-rich secretion produced in the crop — for the first week, then gradually introduce seeds. Young fledge at 12–15 days.

SONG & COMMUNICATION

The Mourning Dove's call is a haunting, hollow coo — typically "ooo-woo-woo-woo" with the first note higher in pitch. It is one of the most widely recognized bird sounds in North America and carries long distances in still air. The call is given year-round but peaks during the breeding season. The loud wing whistle produced at takeoff serves as an alarm signal, alerting nearby birds to a predator. Males also give a soft courtship coo during pair bonding.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With an estimated 350 million individuals, the Mourning Dove is the most abundant game bird and one of the most numerous wild birds in North America. It is legally hunted in 41 U.S. states, with an annual harvest exceeding 20 million birds — yet the population remains stable thanks to its extraordinary reproductive rate. It has benefited from the expansion of agriculture and suburban habitats across the continent.

FEEDER TIPS

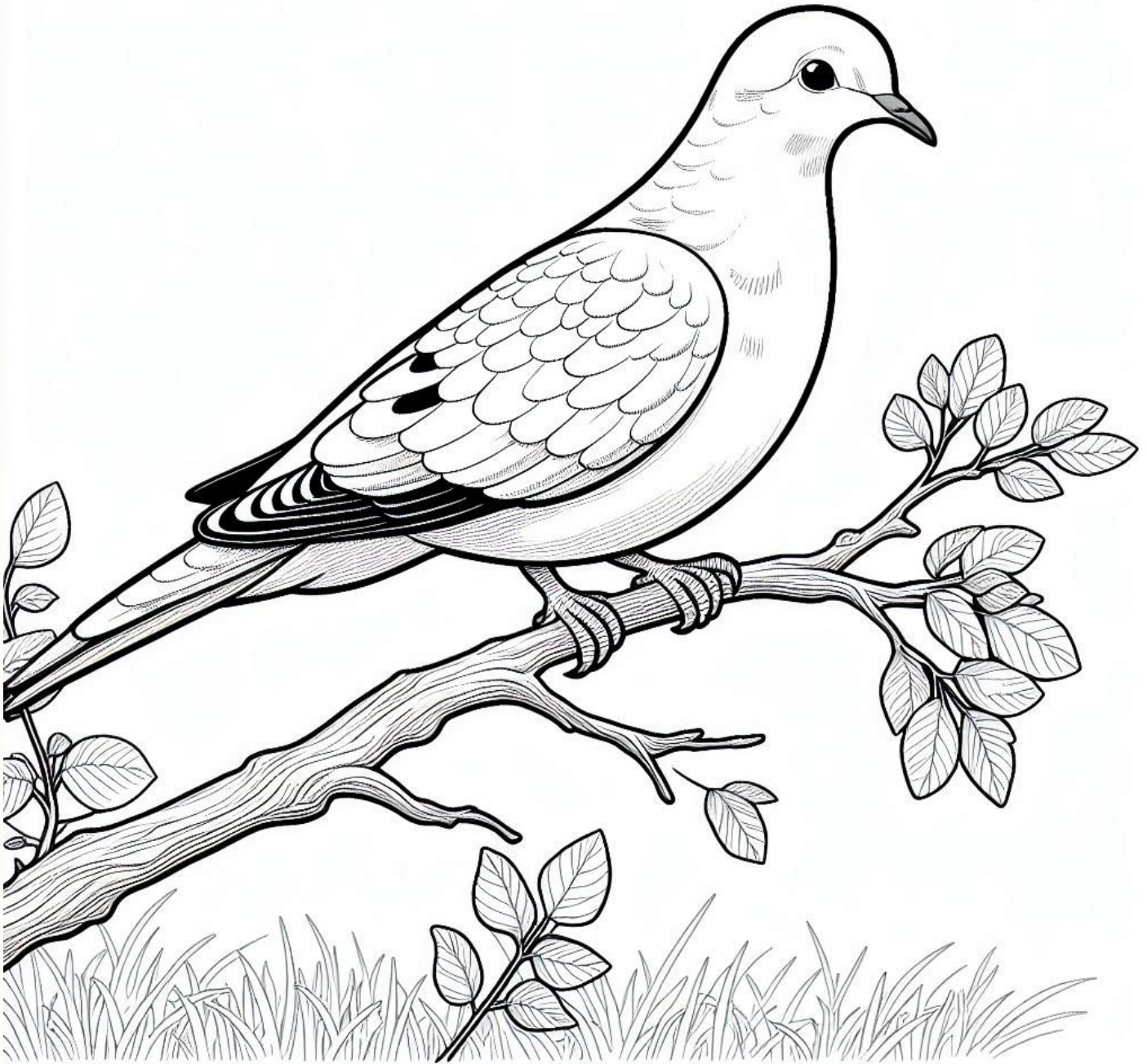
- Scatter white millet and cracked corn directly on the ground
- A large, flat ground tray or platform feeder suits them best
- Mourning Doves arrive at feeders early in the morning and at dusk
- Keep the feeding area clear of deep snow — they cannot dig through it
- They are skittish — minimize foot traffic near the feeding area

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Body	Warm buffy tan — soft and uniform
Wings	Light brown with black spots
Neck patch	Iridescent blue-green (male)
Tail	Long, pointed; white-tipped edges
Bill	Short, slender, dark gray
Legs & feet	Red-pink

COLOR ME!

Mourning Dove • *Zenaida macroura* • Module 13



MODULE 14 / 40

Red-bellied Woodpecker

*Melanerpes carolinus***QUICK STATS**

Length	9.4 in (24 cm)
Wingspan	13.0 – 16.5 in (33 – 42 cm)
Weight	2.0 – 3.2 oz (56 – 91 g)
Lifespan	Up to 12 years (avg. 2–3 in wild)

PRESENTATION

Despite its name, the Red-bellied Woodpecker's most striking feature is its brilliant red cap — not its belly, which shows only a faint reddish wash that is rarely visible in the field. This bold, vocal woodpecker has expanded dramatically northward over the past century, colonizing new areas of the northeastern United States and southern Canada as forests have matured. With its ladder-backed black-and-white barring and loud, rolling "churr" call, it is one of the most conspicuous and entertaining birds at suet feeders across eastern North America. Its remarkably long, barbed tongue — extending up to 2 inches beyond the bill — allows it to extract insects from deep bark crevices and remove seeds from tight spaces.

HABITAT & RANGE

The Red-bellied Woodpecker is found year-round across the eastern United States, from southern Canada south to Florida and west to the Great Plains. It inhabits mature deciduous and mixed forests, forest edges, suburban parks, and wooded gardens with large trees. It shows a preference for oak-dominated forests where acorns are plentiful, but readily adapts to suburban environments wherever there are large, established trees. It is a permanent resident throughout its range and does not migrate.

DIET & FEEDING

- Suet cakes — the top feeder attraction for this species
- Black-oil sunflower seeds and sunflower chips
- Whole and shelled peanuts
- Acorns and beechnuts (cached in bark crevices)
- Insects and larvae extracted from dead wood
- Wild berries and small fruits — especially in autumn

BEHAVIOR & SOCIAL LIFE

Red-bellied Woodpeckers are bold, assertive birds at feeders, often dominating smaller species such as Downy Woodpeckers and nuthatches. They are year-round residents that maintain fixed territories, defended loudly with calls and drumming. Like other woodpeckers, they are avid food cachers — storing seeds, acorns, and insects in bark crevices, post holes, and even the gaps in wooden siding. Their long, sticky tongue is specially adapted to extract prey from deep crevices, and their stiff tail feathers act as a prop against tree trunks. They often join mixed foraging flocks with chickadees and nuthatches in winter.

NESTING & BREEDING

Red-bellied Woodpeckers excavate nest cavities in dead or dying trees, typically at heights of 5–70 feet. The male does most of the excavation, taking 1–2 weeks to complete the cavity. Both parents incubate the 4–5 white eggs for 12 days. Both parents feed the nestlings, which fledge at 24–27 days. The species raises 2–3 broods per season. Old cavities are frequently reused in subsequent years and provide critical nesting sites for secondary cavity nesters such as Eastern Bluebirds and screech-owls.

SONG & COMMUNICATION

The Red-bellied Woodpecker is one of the most vocal woodpeckers in eastern North America. Its primary call is a loud, rolling "churr" or "kwirr" that carries clearly through the forest. It also gives a sharp "chek" alarm call and a variety of softer conversational notes between mates. Drumming on resonant dead wood or metal surfaces serves as a territorial advertisement, with rapid, sustained bursts that can be heard from considerable distances.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Increasing
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The Red-bellied Woodpecker has been one of the great conservation success stories of recent decades. Once largely confined to the southeastern United States, it has expanded its range steadily northward since the mid-20th century, benefiting from forest maturation, milder winters, and the proliferation of backyard feeders. Its population is estimated at approximately 16 million individuals and continues to grow.

FEEDER TIPS

- Mount a suet cage directly on a tree trunk — ideal positioning for this species
- Offer whole peanuts in a mesh peanut feeder
- Large hopper or platform feeders with sunflower seeds attract them readily
- Install a nest box with a 2-inch entrance hole on a large dead snag
- Red-bellied Woodpeckers cache food — expect them to make repeated quick visits

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown (male)	Full red cap — forehead to nape
Crown (female)	Red on nape only; gray forehead
Back & wings	Bold black-and-white barring
Face & underparts	Pale gray-buff
Belly patch	Faint red-orange wash (often hidden)
Bill	Long, chisel-tipped, dark gray

COLOR ME!

Red-bellied Woodpecker • *Melanerpes carolinus* • Module 14



MODULE 15 / 40

Tufted Titmouse

Baeolophus bicolor

QUICK STATS

Length	5.5 – 6.3 in (14 – 16 cm)
Wingspan	7.9 – 10.2 in (20 – 26 cm)
Weight	0.6 – 0.9 oz (18 – 26 g)
Lifespan	Up to 13 years (avg. 2 in wild)

PRESENTATION

The Tufted Titmouse is one of the most charming and acrobatic visitors to eastern feeders — a small, perky bird with a distinctive gray crest, large dark eyes, and a bold black forehead patch that gives it a perpetually alert, wide-eyed expression. It is closely related to the chickadees and shares many of their bold, inquisitive behaviors. Like the Black-capped Chickadee, it readily comes to feeders and is bold enough to take seeds from an outstretched hand with patience. Its loud, whistled song — a ringing "peter-peter-peter" repeated insistently — is one of the most recognizable sounds of eastern woodlands from late winter onward.

HABITAT & RANGE

The Tufted Titmouse is a permanent resident of the eastern United States, from the Great Plains east to the Atlantic coast, and from southern Ontario and New England south to Florida and the Gulf Coast. Like the Red-bellied Woodpecker, it has expanded its range significantly northward over the past 50 years, aided by forest maturation, milder winters, and backyard feeders. It inhabits mature deciduous and mixed forests, forest edges, orchards, parks, and wooded suburban gardens. It shows a strong preference for areas with large oaks and other mast-producing trees.

DIET & FEEDING

- Black-oil sunflower seeds — the clear favorite at feeders
- Sunflower chips and safflower seeds
- Suet cakes — especially valued in winter
- Whole and shelled peanuts
- Acorns, beechnuts, and hickory nuts (cached for winter)
- Insects, caterpillars, and spiders — critical in breeding season

BEHAVIOR & SOCIAL LIFE

Tufted Titmice are bold, curious, and highly social birds. Outside the breeding season they join mixed foraging flocks with Black-capped Chickadees, Downy Woodpeckers, and White-breasted Nuthatches, often serving as the "nuclear" species around which the flock organizes. They are dominant over chickadees at feeders, regularly displacing them for the best seeds. Like chickadees, they hammer seeds open by holding them with their feet and striking with the bill. Titmice are avid food cachers, storing seeds and nuts in bark crevices, soil, and leaf litter within their territory for retrieval during winter cold snaps.

NESTING & BREEDING

Tufted Titmice are cavity nesters, using natural holes in trees, old woodpecker cavities, and nest boxes. The female builds a bulky cup nest inside the cavity using leaves, moss, bark strips, and an inner lining of soft material — notably including hair plucked directly from living animals (mammals and even humans). Clutches of 5–6 white eggs with reddish-brown spots are incubated by the female for 13–14 days. Both parents feed the nestlings, which fledge at 15–16 days. Offspring from the first brood sometimes remain as helpers to assist with subsequent broods — a rare behavior in North American songbirds.

SONG & COMMUNICATION

The Tufted Titmouse's song is a loud, clear, whistled "peter-peter-peter" or "here-here-here" repeated 4–11 times — one of the earliest and most persistent bird songs of late winter, often beginning in January on warm days. It also shares a vocabulary of scolding "dee" calls with chickadees, and the two species understand and respond to each other's alarm calls. A variety of harsh, churring scold notes are given when a predator is detected.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Increasing
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The Tufted Titmouse population is estimated at approximately 8 million individuals and has been increasing steadily, driven by northward range expansion. It benefits strongly from backyard feeders, which have allowed it to survive harsher winters and colonize regions where it was formerly absent. Retaining mature trees with natural cavities and installing nest boxes are effective ways to support local populations.

FEEDER TIPS

- Black-oil sunflower seeds in a tube or hopper feeder are ideal
- Offer a suet cage near large trees — titmice use it year-round
- Shelled peanuts in a mesh feeder attract them reliably
- Install a nest box (1.25-inch entrance) on a tree — they will use it
- Titmice are among the boldest feeder birds — hand-feeding is achievable with patience

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crest	Gray — tall and pointed
Forehead	Black patch just above bill
Back & wings	Medium gray
Face & throat	White to pale gray
Flanks	Warm peach-orange wash
Bill	Short, stout, dark gray

COLOR ME!

Tufted Titmouse • *Baeolophus bicolor* • Module 15



MODULE 16 / 40

Carolina Wren

Thryothorus ludovicianus

QUICK STATS

Length	4.7 – 5.5 in (12 – 14 cm)
Wingspan	11.4 in (29 cm)
Weight	0.6 – 0.8 oz (18 – 22 g)
Lifespan	Up to 6 years (avg. 2 in wild)

PRESENTATION

The Carolina Wren is one of the most surprising birds in the backyard — a tiny, round-bodied bird with a cocked tail and a bold white eyebrow stripe that produces one of the loudest songs of any North American bird relative to its size. Its rich, warm rufous plumage, perpetually upturned tail, and restless, energetic movements make it instantly endearing. Unlike most wrens, the Carolina Wren is a year-round resident that forms permanent pair bonds, with mates staying in close contact throughout the year. Pairs duet frequently — the male belting out his ringing "teakettle-teakettle-teakettle" while the female responds with a dry chatter — a remarkable coordination between mates.

HABITAT & RANGE

The Carolina Wren is found year-round across the eastern United States, from Nebraska and southern Iowa east to the Atlantic coast, and from southern Ontario south through Florida and into Central America. It inhabits dense, brushy undergrowth in deciduous and mixed forests, forest edges, thickets, vine tangles, brushy gardens, and overgrown fields. It thrives in suburban and rural gardens with dense shrubs, brush piles, and woody debris. It is highly sensitive to cold winters — severe ice storms can cause dramatic local population crashes, from which recovery takes several years.

DIET & FEEDING

- Suet cakes — the top feeder food for Carolina Wrens
- Peanut butter and peanut pieces
- Mealworms — live or dried, eagerly taken
- Insects and spiders — the bulk of the natural diet year-round
- Small berries and seeds occasionally in winter
- Bark butter (peanut butter-fat mixture) spread on tree trunks

BEHAVIOR & SOCIAL LIFE

Carolina Wrens are intensely curious and remarkably bold for their size. They investigate every nook and cranny of their territory — peering into flowerpots, exploring open garages and porches, and probing bark and leaf litter for insects with their long, curved bills. Unlike many songbirds, Carolina Wrens maintain their pair bond year-round. The male and female remain in constant vocal contact, often duetting together. They are non-migratory and fiercely territorial, defending their home range against other wrens with loud singing and aggressive chases. They roost communally in cold weather, sometimes piling into nest boxes or dense vegetation to conserve warmth.

NESTING & BREEDING

Carolina Wrens are famously eclectic nesters, placing their domed cup nests in almost any sheltered cavity — natural tree holes, old woodpecker cavities, nest boxes, flowerpots, hanging baskets, boots left on a porch, or gaps in outbuildings. The male builds several "dummy" nests that are never used, while the female selects and completes the final nest. Clutches of 4–6 white eggs with reddish-brown speckles are incubated by the female for 12–16 days. Both parents feed the nestlings, which fledge at 12–14 days. Pairs raise 2–3 broods per season.

SONG & COMMUNICATION

The Carolina Wren's song is extraordinarily loud for such a small bird — a clear, ringing "teakettle-teakettle-teakettle" or "cheery-cheery-cheery" that carries 100 meters or more. Males have a repertoire of 27–40 distinct song types and switch between them constantly. Females respond with a buzzy chatter, creating antiphonal duets. Both sexes give a variety of sharp "jimp" and churring alarm calls when disturbed. Song is heard year-round, with intensity peaking in early spring.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Increasing
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The Carolina Wren population is estimated at approximately 14 million individuals and has been expanding northward over recent decades, aided by warming winters. However, it remains vulnerable to severe cold snaps and ice storms — a single harsh winter can eliminate local populations. Providing dense shrub cover, brush piles, suet, and nest boxes are among the most effective ways to support this species through winter.

FEEDER TIPS

- Suet cakes near dense shrubs are the most reliable attraction
- Offer live or dried mealworms in a shallow dish — they adore them
- Brush piles and log piles near the feeder provide essential cover
- A low platform feeder with peanut pieces works well
- Carolina Wrens nest in boxes — a 1.5-inch entrance box near cover often works

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown & back	Rich warm brown — rufous tones
Wings & tail	Brown with fine dark barring
Supercilium	Bold white eyebrow stripe
Throat & breast	Buffy white to warm buff
Flanks & belly	Deeper buffy-orange wash
Bill	Long, slender, curved, dark

COLOR ME!

Carolina Wren • *Thryothorus ludovicianus* • Module 16



MODULE 17 / 40

American Crow

*Corvus brachyrhynchos***QUICK STATS**

Length	17.5 – 21.0 in (44 – 53 cm)
Wingspan	33.5 – 39.4 in (85 – 100 cm)
Weight	11.2 – 21.9 oz (316 – 620 g)
Lifespan	Up to 16 years (avg. 7–8 in wild)

PRESENTATION

The American Crow is one of the most intelligent, adaptable, and ecologically important birds in North America — a large, all-black bird with a heavy bill, fan-shaped tail, and an extraordinary capacity for learning, problem-solving, and social organization. Crows can recognize individual human faces, use tools, pass information between generations, and hold "funerals" for fallen flock members — behaviors once thought exclusive to primates. Their loud, familiar "caw" is one of the most iconic sounds of the North American landscape. Highly social and opportunistic, they thrive in virtually every habitat from wilderness to city centers.

HABITAT & RANGE

The American Crow is found year-round across most of North America, from southern Canada through the entire contiguous United States except for the driest desert regions. It occupies an extraordinary range of habitats: farmland, forest edges, river valleys, coastal shorelines, suburban neighborhoods, parks, and city centers. It is one of the few species that has benefited from human landscape changes, thriving wherever there is a mix of open foraging areas and trees for roosting and nesting. Northern populations move south in winter, often forming massive communal roosts of thousands to hundreds of thousands of birds.

DIET & FEEDING

- Virtually omnivorous — one of the broadest diets of any bird
- Grain, corn, and seeds scavenged from fields and feeders
- Insects, earthworms, and small mammals
- Carrion and roadkill — an important ecological scavenger
- Eggs and nestlings of other birds
- Fruit, berries, nuts, and human food scraps

BEHAVIOR & SOCIAL LIFE

American Crows are among the most socially complex birds in the world. They live in extended family groups where offspring from previous years help their parents raise subsequent broods — a behavior called cooperative breeding. Family groups maintain tight bonds year-round, foraging together, mobbing predators collectively, and communicating with a rich vocabulary of calls. Crows can recognize and remember individual human faces for years, and will "hold grudges" against people who have threatened them while tolerating those who have been kind. In winter, unrelated crows gather in communal roosts that can number in the millions, providing warmth and safety in numbers.

NESTING & BREEDING

Crows nest in late winter and early spring, building large, bulky cup nests of sticks lined with bark strips, grass, and soft material in the crotch of a tall tree. Clutches of 3–6 pale green or blue-gray eggs with brown spots are incubated by the female for 17–19 days. Both parents and often "helper" offspring from previous years feed the nestlings. Young fledge at 28–35 days but remain with the family group for 1–5 years. Breeding pairs mate for life.

SONG & COMMUNICATION

The American Crow has one of the most complex vocal repertoires of any North American bird — researchers have identified over 20 distinct call types. The familiar "caw" varies in pitch, repetition, and rhythm to convey different messages: alarm, aggression, assembly, and contact. Crows also produce a remarkable array of rattles, coos, clicks, and mimicked sounds. Family members recognize each other's voices individually.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The American Crow population is estimated at 27 million individuals and is broadly stable. It suffered dramatic declines during the initial spread of West Nile Virus in the early 2000s — crows are among the most susceptible species — but has largely recovered in most areas. It remains one of the most abundant and visible large birds in North America.

FEEDER TIPS

- Crows rarely use traditional feeders — offer food on the ground or a large platform
- Whole corn, peanuts in shell, and kitchen scraps attract them reliably
- They are wary — place food in an open area where they can watch for danger
- Crows remember generous feeding spots and return reliably for years
- Avoid feeding near songbird feeders — crows can intimidate smaller birds

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Entire body	Jet black — uniform and glossy
Iridescence	Blue-violet sheen in sunlight
Bill	Heavy, curved, black
Legs & feet	Black
Eye	Dark brown — appears black
Juvenile plumage	Duller black; blue eye at first

COLOR ME!

American Crow • *Corvus brachyrhynchos* • Module 17



MODULE 18 / 40

Red-winged Blackbird

*Agelaius phoeniceus***QUICK STATS**

Length	6.7 – 9.1 in (17 – 23 cm)
Wingspan	12.2 – 15.8 in (31 – 40 cm)
Weight	1.1 – 2.7 oz (32 – 77 g)
Lifespan	Up to 15 years (avg. 2 in wild)

PRESENTATION

The Red-winged Blackbird is one of the most abundant birds in North America — and among the most instantly recognizable. The male's jet-black plumage set against blazing scarlet-and-yellow shoulder patches is one of the most vivid color combinations in North American ornithology. Few sights signal the arrival of spring more dramatically than a male Red-winged Blackbird perched atop a cattail, wings half-spread to display those brilliant epaulets, delivering his liquid, bubbling "conk-la-reeee" call. In contrast, females are so heavily streaked and brown that beginners often fail to recognize them as the same species.

HABITAT & RANGE

The Red-winged Blackbird breeds across virtually all of North America, from Alaska and Canada south through the United States, Mexico, and into Central America. It is one of the most widespread breeding birds on the continent. It shows a strong preference for wetland habitats — cattail marshes, reed beds, wet meadows, and the edges of ponds and lakes — but also nests in dry upland fields and roadsides. In winter, northern populations migrate south and form enormous mixed flocks with other blackbirds and starlings, sometimes numbering in the millions.

DIET & FEEDING

- Black-oil sunflower seeds and cracked corn at feeders
- White millet scattered on the ground
- Wild grass seeds, grain, and weed seeds — the core of the natural diet
- Insects, spiders, and aquatic invertebrates during breeding season
- Wild berries and small fruits in autumn
- Waste grain gleaned from agricultural fields in winter

BEHAVIOR & SOCIAL LIFE

Red-winged Blackbirds are highly polygynous — a single male may hold a territory with up to 15 females nesting within it, though 2–3 is more typical. Males arrive on the breeding grounds weeks before females and spend much of the early season in fierce territorial battles, displaying their red shoulder patches and singing constantly from prominent perches. They are remarkably aggressive defenders of their territories, regularly dive-bombing much larger animals — hawks, herons, crows, and even humans — that venture too close to active nests. Outside the breeding season they are highly gregarious, forming enormous mixed-species flocks that roost communally.

NESTING & BREEDING

Females build deep, woven cup nests of grasses and aquatic vegetation, lashed to cattail stems or low shrubs over or near water. Clutches of 3–4 pale blue-green eggs with dark scrawls are incubated by the female alone for 11–13 days. Both parents feed the nestlings, which fledge at 11–14 days. Pairs raise 2–3 broods per season. Females accept cowbird eggs at a high rate, making them frequent brood parasite hosts.

SONG & COMMUNICATION

The Red-winged Blackbird's song is one of the most evocative sounds of North American wetlands — a liquid, gurgling "conk-la-reeee" or "oak-a-lee" delivered with a dramatic wing-spread display. Males have individual song repertoires used to advertise territory. The sharp "check" call and harsh chattering alarm notes are given by both sexes. Females produce a scolding chatter when disturbed near the nest.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining
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Despite being one of the most abundant birds in North America — with a population estimated at 150 million individuals — the Red-winged Blackbird has declined significantly since 1970, losing an estimated 30% of its population. Wetland drainage, agricultural intensification, and the loss of native grasslands are the primary drivers. Protecting and restoring wetland habitats is critical for this species.

FEEDER TIPS

- Scatter cracked corn and white millet on a large ground platform
- Red-winged Blackbirds arrive in spring flocks — expect multiple birds at once
- They prefer open feeding areas with water or marsh nearby
- In winter, mixed blackbird flocks may descend on feeders by the hundreds
- Planting native cattails and marsh plants near water attracts nesting pairs

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Body (male)	Glossy jet black
Shoulder patch (male)	Brilliant scarlet-red
Shoulder border	Yellow to pale yellow
Female — overall	Dark brown, heavily streaked
Female — face	Pale buff supercilium stripe
Bill	Sharp, pointed, black

COLOR ME!

Red-winged Blackbird • *Agelaius phoeniceus* • Module 18



MODULE 19 / 40

Baltimore Oriole

*Icterus galbula***QUICK STATS**

Length	6.7 – 7.5 in (17 – 19 cm)
Wingspan	9.1 – 11.8 in (23 – 30 cm)
Weight	1.1 – 1.4 oz (30 – 40 g)
Lifespan	Up to 11 years (avg. 6 in wild)

PRESENTATION

The Baltimore Oriole is one of the most spectacular and eagerly awaited migrants in eastern North America — the male's blazing flame-orange and jet-black plumage is rivaled by few birds on the continent. Named after the colors of Lord Baltimore's coat of arms, it is the state bird of Maryland and a beloved symbol of spring. Its arrival at feeders in late April and May — often announced by its rich, fluty, whistled song from the canopy of an elm or cottonwood — is one of the most anticipated events of the backyard birding year. It is uniquely attracted to halved oranges and grape jelly offered at specialized feeders.

HABITAT & RANGE

The Baltimore Oriole breeds across the eastern United States and southern Canada, from the Great Plains east to the Atlantic coast and from southern Canada south to northern Georgia and Oklahoma. It is a long-distance migrant, wintering from southern Mexico south through Central America to northern South America. It inhabits open deciduous and mixed woodland, forest edges, riparian corridors, orchards, parks, and large suburban gardens with tall trees — particularly elms, cottonwoods, and maples. It nests high in the canopy of large trees.

DIET & FEEDING

- Halved oranges skewered on a feeder spike — irresistible to orioles
- Grape jelly in a small shallow dish — a top feeder attraction
- Nectar feeders (oriole-style, with larger ports)
- Ripe dark berries: mulberries, serviceberries, cherries
- Insects and caterpillars — the core of the breeding-season diet
- Flower nectar probed directly from blossoms

BEHAVIOR & SOCIAL LIFE

Baltimore Orioles are solitary and territorial during the breeding season, with males defending territories through song and aerial chases. They are canopy specialists, spending most of their time high in the crowns of tall trees where they are often heard before they are seen. Despite their shyness in the treetops, males readily come down to feeder stations offering oranges and jelly. They are migratory, departing for the tropics as early as late July — among the earliest fall migrants of any North American songbird. Fall birds show more subdued plumage than spring males.

NESTING & BREEDING

The Baltimore Oriole builds one of the most remarkable nests of any North American bird — a deep, sock-shaped pouch woven from plant fibers, grasses, and string, suspended from the tip of a high, drooping branch. The female does all nest construction, weaving the structure over 1–2 weeks. Clutches of 4–5 pale grayish eggs with dark scrawls are incubated by the female for 12–14 days. Both parents feed the nestlings, which fledge at 12–14 days. One brood per season.

SONG & COMMUNICATION

The Baltimore Oriole's song is a series of rich, fluty whistles — clear, melodious phrases that carry well from the treetops. Each male has a distinctive song with regional dialects. The call is a dry, rattling chatter and a clear whistled "hew-li" contact note. Females also sing — their song is shorter but similarly rich in tone. Song is heard primarily from arrival in late April through mid-summer.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining
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The Baltimore Oriole population is estimated at 12 million individuals but has declined by approximately 24% since 1970. Loss of large elms to Dutch Elm Disease removed a favored nesting tree across much of its range. Additional threats include habitat loss on wintering grounds, pesticide use reducing insect availability, and window collisions. Planting native berry-producing trees and offering feeder food in spring are meaningful ways to support this species.

FEEDER TIPS

- Put out halved oranges and grape jelly by May 1 — before arrival
- Use a dedicated oriole feeder with orange-colored ports and fruit spikes
- Place feeders in the open near tall trees — orioles need clear flight paths
- Replace jelly and fruit every 1–2 days in warm weather to prevent spoilage
- Plant native serviceberries and mulberries — natural magnets for orioles

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head & back (male)	Deep black
Breast & belly (male)	Brilliant flame orange
Wings (male)	Black with white wing bar
Female — overall	Olive-yellow to dull orange
Female — wings	Dark brown with two wing bars
Bill	Long, pointed, blue-gray

COLOR ME!

Baltimore Oriole • *Icterus galbula* • Module 19



MODULE 20 / 40

Cedar Waxwing

Bombusilla cedrorum

QUICK STATS

Length	5.5 – 6.7 in (14 – 17 cm)
Wingspan	8.7 – 11.8 in (22 – 30 cm)
Weight	1.1 – 1.4 oz (32 – 40 g)
Lifespan	Up to 13 years (avg. 5 in wild)

PRESENTATION

The Cedar Waxwing is perhaps the most elegantly plumaged bird in North America — a sleek, silky bird of extraordinary refinement, with smooth blended plumage in tones of warm brown, gray, and pale yellow, set off by a bold black mask, a brilliant yellow tail-tip band, and the mysterious red waxy droplets on its wing tips that give the species its name. These red tips are actually extensions of the feather shafts and increase in number with age — older birds have more, making them more attractive to potential mates. Cedar Waxwings are nomadic flock birds, appearing unpredictably wherever berries are abundant and stripping entire trees in a matter of hours.

HABITAT & RANGE

The Cedar Waxwing breeds across southern Canada and the northern United States, from British Columbia to Newfoundland and south through the Appalachians and northern Great Plains. In winter it roams widely across the entire contiguous United States and into Central America, following berry crops. It inhabits open woodlands, forest edges, orchards, suburban parks, and gardens — wherever fruiting trees and shrubs are present. It shows a particularly strong association with fruiting cedars, serviceberries, crabapples, and mountain ash.

DIET & FEEDING

- Wild berries — cedar, serviceberry, crabapple, hawthorn, holly
- Mountain ash berries — a winter favorite
- Dogwood, viburnum, and elderberry fruits
- Insects caught in aerial sallies during summer — especially near water
- Cherry and apple fruits — sometimes causing intoxication from fermentation
- Flower petals occasionally consumed in spring

BEHAVIOR & SOCIAL LIFE

Cedar Waxwings are among the most social songbirds in North America, living in tight flocks year-round and rarely seen alone. Flocks move nomadically across the landscape, locating fruiting trees by sight and sound and descending en masse to strip them. They show a remarkable behavior of passing berries or flower petals beak-to-beak along a perched row of birds — an apparent bonding or courtship ritual. Unlike most songbirds, waxwings do not defend feeding territories and tolerate conspecifics freely at food sources. They are highly efficient berry consumers, capable of eating their body weight in fruit daily, and are important seed dispersers for many native plants.

NESTING & BREEDING

Cedar Waxwings are late nesters, typically not beginning until June or July to coincide with the peak availability of summer berries. They often nest in loose colonies. The female builds a bulky cup nest of grasses, twigs, and plant fibers in a tree fork, lined with fine grasses and hair. Clutches of 3–5 pale gray eggs with dark spots are incubated by the female for 11–13 days. Both parents feed the nestlings, which fledge at 14–18 days. One to two broods per season.

SONG & COMMUNICATION

The Cedar Waxwing's vocalizations are among the most distinctive in North American birding — a high, thin, sibilant trill or whistle, often written as "seeeee" or "srrrrr," that is easily missed by those with high-frequency hearing loss. Flocks produce a constant soft chorus of these calls as they feed — often the first indication that waxwings are present overhead. They have no true song in the traditional sense.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Increasing
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The Cedar Waxwing population is estimated at 52 million individuals and has increased significantly since the 1960s, benefiting from the maturation of forest edges and the proliferation of ornamental fruiting trees in suburban landscapes. It is one of the few fruit-specialist songbirds to have thrived in human-modified environments. Window collisions are a significant mortality source for this species.

FEEDER TIPS

- Plant native serviceberry, crabapple, hawthorn, and holly — they don't use traditional feeders
- Mountain ash is one of the most reliable trees for attracting winter flocks
- Leave fruiting shrubs unpruned through winter — waxwings will find them
- A shallow birdbath near fruiting trees greatly increases attractiveness
- Listen for the high, thin "see" call overhead — often the first sign of a flock

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head & crest	<i>Silky warm brown — tawny</i>
Black mask	<i>Bold black eye mask — both sexes</i>
Back & breast	<i>Smooth gray-brown, blending</i>
Belly	<i>Pale yellow fading to white</i>
Tail tip	<i>Bright yellow band</i>
Wing tips	<i>Red waxy droplets (adults)</i>

COLOR ME!

Cedar Waxwing • *Bombycilla cedrorum* • Module 20



MODULE 21 / 40

House Finch

Haemorrhous mexicanus

QUICK STATS

Length	5.1 – 5.5 in (13 – 14 cm)
Wingspan	7.9 – 9.8 in (20 – 25 cm)
Weight	0.6 – 0.9 oz (16 – 27 g)
Lifespan	Up to 11 years (avg. 2 in wild)

PRESENTATION

The House Finch is one of the most familiar and abundant feeder birds in North America — a cheerful, sociable finch whose rosy-red male is a year-round fixture at sunflower feeders from coast to coast. Native to the western United States and Mexico, the House Finch was illegally sold as a cage bird (marketed as "Hollywood Finches") in New York in the 1940s. When dealers faced prosecution under the Migratory Bird Act, they released their birds on Long Island — and from that small population, the species colonized the entire eastern United States within 50 years, one of the most remarkable range expansions in North American ornithological history. Today it is among the top five most abundant feeder birds nationally.

HABITAT & RANGE

The House Finch is now found year-round across virtually the entire contiguous United States, southern Canada, and Mexico. It thrives in human-modified landscapes: suburban gardens, parks, farmland, orchards, and city centers. It is rarely found in dense forest or wilderness, strongly preferring areas with buildings, feeders, and ornamental plantings. It is a permanent resident throughout its range and does not migrate, though local movements occur in response to food availability.

DIET & FEEDING

- Black-oil sunflower seeds — the top feeder choice
- Nyjer (thistle) seed — eagerly taken from tube feeders
- Safflower seeds and sunflower chips
- Wild weed seeds: dandelion, thistle, mustard, knotweed
- Wild berries and small fruits — especially in autumn
- Flower buds and petals in spring

BEHAVIOR & SOCIAL LIFE

House Finches are highly gregarious, forming flocks of up to several hundred birds outside the breeding season. They are boisterous and active at feeders, often arriving in chattering waves and dominating smaller birds. The intensity of the male's red coloration is entirely diet-dependent — birds with access to carotenoid-rich foods such as berries and fruits develop deeper, more saturated reds, while those on poor diets may be pale orange or even yellowish. Females use this color intensity as a reliable indicator of male quality when choosing a mate. House Finches are highly adaptable and frequently nest on buildings, in hanging baskets, and in wreaths.

NESTING & BREEDING

House Finches are prolific and flexible nesters. They build compact cup nests of grasses, leaves, and rootlets in a wide variety of sites — tree cavities, dense shrubs, building ledges, hanging planters, and even old nests of other species. The female incubates 2–6 pale blue eggs with fine dark spots for 13–14 days. The male feeds the female during incubation. Both parents feed the nestlings regurgitated seeds, which fledge at 12–19 days. Pairs raise 2–3 broods per season, sometimes beginning in February in southern areas.

SONG & COMMUNICATION

The House Finch's song is a bright, rambling warble of musical phrases — cheerful, variable, and delivered with great energy from a prominent perch. Males sing year-round, including on cold winter days. The call is a sharp "wheet" or rising "wiit." House Finches are one of the most frequent singers at winter feeders, often heard before they are seen.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining slightly
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The House Finch population is estimated at 267 million individuals but has declined noticeably since the mid-1990s, largely due to an epidemic of Mycoplasmal conjunctivitis — a bacterial eye disease that spread rapidly through feeder populations beginning in 1994. Keeping feeders scrupulously clean is the single most important action feeder owners can take to protect House Finch populations.

FEEDER TIPS

- Black-oil sunflower seeds in a tube feeder are the top attraction
- Clean feeders thoroughly every 1–2 weeks — House Finches are prone to eye disease
- Remove and discard any birds seen with swollen, crusty eyes immediately
- Nyjer feeders bring in mixed flocks of House Finches and goldfinches
- House Finches nest on buildings — a hanging planter may become a nest site

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head & breast (male)	<i>Rosy red — intensity varies by diet</i>
Rump (male)	<i>Bright red — key field mark</i>
Back & wings	<i>Brown, streaked dark</i>
Belly	<i>White with brown streaking</i>
Female — overall	<i>Brown, heavily streaked — no red</i>
Bill	<i>Short, curved, conical, pale</i>

COLOR ME!

House Finch • *Haemorhous mexicanus* • Module 21



MODULE 22 / 40

Pine Siskin

Spinus pinus

QUICK STATS

Length	4.3 – 5.5 in (11 – 14 cm)
Wingspan	7.1 – 8.7 in (18 – 22 cm)
Weight	0.4 – 0.6 oz (12 – 18 g)
Lifespan	Up to 9 years (avg. 2 in wild)

PRESENTATION

The Pine Siskin is a small, heavily streaked finch of the boreal and montane forests — an irruptive winter visitor whose appearances at feeders are excitingly unpredictable. In years when conifer seed crops fail across the North, Pine Siskins erupt southward in enormous numbers, descending on nyjer feeders and birch trees across the continent. In other years they are virtually absent. The subtle yellow flash in the wings and at the base of the tail is the key field mark that separates this species from other streaked finches. Despite its plain appearance, the Pine Siskin is a remarkably hardy bird, capable of surviving temperatures as low as -94°F (-70°C) by raising its metabolic rate.

HABITAT & RANGE

The Pine Siskin breeds across the boreal forests of Canada and Alaska, and in montane conifer forests south through the Rockies, Cascades, and Sierra Nevada into Mexico. In winter it moves south and east in variable numbers, potentially appearing anywhere in the contiguous United States during irruption years. It inhabits coniferous and mixed forests, forest edges, weedy fields, and alder and birch stands. In irruption years it frequently visits nyjer feeders in suburban gardens far outside its normal range.

DIET & FEEDING

- Nyjer (thistle) seed — the top feeder choice, attracts large flocks
- Black-oil sunflower seeds and sunflower chips
- Conifer seeds extracted from open cones — the core wild food
- Alder and birch catkin seeds in late winter
- Weed seeds: thistle, dandelion, ragweed
- Salt and mineral deposits — regularly visits salt licks and road salt

BEHAVIOR & SOCIAL LIFE

Pine Siskins are intensely gregarious, almost always found in flocks — sometimes mixed with American Goldfinches, Common Redpolls, and other small finches. They are acrobatic feeders, clinging to the tips of conifer branches and hanging upside-down from seed heads with ease. At feeders they can be surprisingly aggressive for their size, using a threatening wing-spread display to dominate larger birds. During irruption years they may appear at feeders by the dozen or even the hundred. They have a remarkable ability to cache seeds and to increase their food intake dramatically before cold nights.

NESTING & BREEDING

Pine Siskins breed in loose colonies in coniferous forests. The female builds a well-concealed shallow cup nest of twigs, grass, and rootlets on a horizontal conifer branch, lined with fur, feathers, and plant down. Clutches of 3–5 pale blue-green eggs with dark spots are incubated by the female for 13 days, while the male feeds her. Both parents feed the nestlings regurgitated seeds. Young fledge at 14–15 days. Pairs raise 1–2 broods per season, with breeding timed to conifer seed availability.

SONG & COMMUNICATION

The Pine Siskin's calls are highly distinctive — a rising, buzzy "zreeeee" or "skreee" that is unlike any other small finch and carries a characteristic ascending quality. Flocks produce a constant chatter of these calls and short musical twitters. The song is a rambling series of musical and buzzy phrases, similar to an American Goldfinch but harsher and more variable.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining
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The Pine Siskin population is estimated at 35 million individuals but has declined significantly — by roughly 80% — since 1970, making it one of the most steeply declining songbirds in North America. Loss of boreal forest, changes in conifer seed production linked to climate change, and Salmonella outbreaks at contaminated feeders are the primary threats. Keeping nyjer feeders scrupulously clean is critical during irruption years.

FEEDER TIPS

- Stock nyjer feeders before November — irruptions can arrive suddenly
- Clean nyjer feeders weekly — Pine Siskins are highly susceptible to Salmonella
- A thistle sock or fine-mesh nyjer tube is ideal; they cling well
- In irruption years, expect them to arrive with American Goldfinch flocks
- Check wing and tail bases for yellow flash — the key ID mark in mixed flocks

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Overall body	Brown, heavily streaked dark
Yellow wing flash	Pale yellow — base of flight feathers
Yellow tail base	Yellow at base of tail — both sexes
Crown	Dark brown with fine streaking
Underparts	Whitish, streaked brown
Bill	Short, sharply pointed, dark

COLOR ME!

Pine Siskin • *Spinus pinus* • Module 22



MODULE 23 / 40

Chipping Sparrow

*Spizella passerina***QUICK STATS**

Length	4.7 – 5.9 in (12 – 15 cm)
Wingspan	8.3 in (21 cm)
Weight	0.4 – 0.6 oz (11 – 16 g)
Lifespan	Up to 9 years (avg. 2–3 in wild)

PRESENTATION

The Chipping Sparrow is one of the most familiar and confiding small sparrows in North America — a trim, clean-looking bird whose bright rufous cap, bold white eyebrow, and unstreaked gray breast make it one of the easiest sparrows to identify in breeding plumage. It is a common inhabitant of suburban gardens, parks, and open woodlands, often nesting close to human activity and readily visiting feeders scattered with white millet. Its song — a long, mechanical trill on a single pitch — is one of the most familiar sounds of North American spring and summer, frequently confused with the trill of an insect. It is sometimes called the "hair bird" for its habit of lining its nest with animal hair.

HABITAT & RANGE

The Chipping Sparrow breeds across most of North America, from the Yukon and Northwest Territories east to Newfoundland and south through virtually the entire United States into Mexico and Central America. It inhabits open and semi-open environments with scattered trees: suburban gardens, parks, orchards, forest edges, open pine woodlands, and grassy areas with shrubs. It shows a particular affinity for lawns edged with conifers. Northern populations migrate south in winter, when the rufous cap is replaced by a duller, streaked brown crown.

DIET & FEEDING

- White millet — the top feeder food, eagerly taken from the ground
- Black-oil sunflower chips and cracked corn
- Wild grass seeds — the dominant natural food
- Weed seeds: crabgrass, chickweed, smartweed
- Insects and spiders — essential during breeding season
- Primarily forages on the ground beneath feeders and shrubs

BEHAVIOR & SOCIAL LIFE

Chipping Sparrows are tame and approachable birds, often allowing close observation as they forage quietly on lawns and garden paths. During the breeding season males are territorial, singing persistently from a tree or shrub perch. Outside the breeding season they become gregarious, forming loose flocks with other sparrows and juncos. They forage almost exclusively on the ground, moving with quick, darting hops and picking up seeds with precise bill movements. Their small size and unobtrusive manner mean they are often overlooked despite being remarkably common in suburban areas.

NESTING & BREEDING

Chipping Sparrows build compact, neatly woven cup nests of grasses and rootlets, almost always lined with fine animal hair — deer hair, horse hair, or even human hair collected near stables and farms. Nests are placed in shrubs or small conifers, typically 3–10 feet above the ground. The female lays 3–5 pale blue eggs with dark spots and incubates them alone for 11–14 days. Both parents feed the nestlings, which fledge at 9–12 days. Pairs raise 2 broods per season. They are frequent cowbird hosts.

SONG & COMMUNICATION

The Chipping Sparrow's song is a prolonged, rapid, mechanical trill on a single pitch — "chrrrrrrrrrr" — lasting 3–4 seconds and remarkably insect-like in quality. It is one of the commonest sounds of suburban spring mornings. The call is a sharp, high "tsip" — thin and easily overlooked. Males sing persistently from dawn to mid-morning throughout the breeding season.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining
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The Chipping Sparrow population is estimated at 230 million individuals but has declined by roughly 27% since 1970. The primary threats are brood parasitism by Brown-headed Cowbirds, habitat loss through suburban development eliminating open grassy areas with scattered trees, and pesticide use reducing insect availability during the breeding season.

FEEDER TIPS

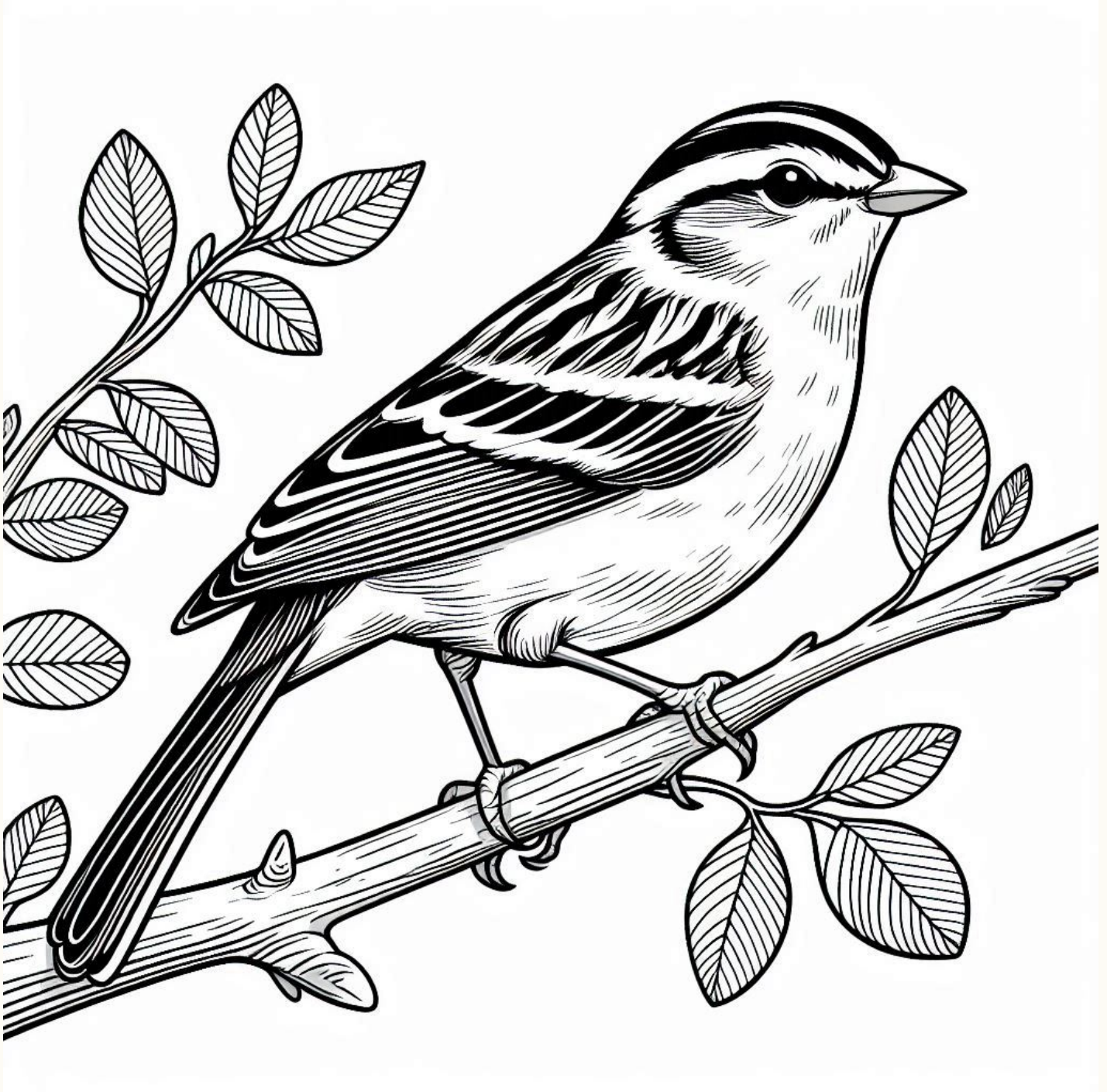
- Scatter white millet on the ground or a low platform feeder
- Chipping Sparrows forage directly beneath tube feeders — no platform needed
- They are tame and often approach within a few feet of a patient observer
- Leave short-mown lawn areas near feeders — they love to forage on grass
- A small birdbath on or near the ground is strongly attractive to them

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown (breeding)	Bright rufous chestnut
White supercilium	Bold white eyebrow stripe
Black eye line	Sharp black line through eye
Back & wings	Brown, streaked dark
Face & underparts	Clean pale gray — unstreaked
Bill	Short, conical, dark gray

COLOR ME!

Chipping Sparrow • *Spizella passerina* • Module 23



MODULE 24 / 40

Eastern Towhee

*Pipilo erythrophthalmus***QUICK STATS**

Length	6.8 – 8.2 in (17 – 21 cm)
Wingspan	7.9 – 11.0 in (20 – 28 cm)
Weight	1.1 – 1.8 oz (32 – 53 g)
Lifespan	Up to 12 years (avg. 3 in wild)

PRESENTATION

The Eastern Towhee is one of the most boldly plumaged and charismatic ground birds of the eastern United States — a large, striking sparrow whose jet-black hood, warm rufous-orange flanks, and clean white belly create one of the most elegant color combinations of any North American songbird. Its vivid red eye is startling at close range. The Eastern Towhee is primarily a bird of dense undergrowth, heard far more often than it is seen — its loud, emphatic "drink-your-teeeeee" song rings from thickets across the eastern woodlands from spring through summer. At feeders it appears mainly in winter, scratching vigorously beneath shrubs for spilled seed.

HABITAT & RANGE

The Eastern Towhee is found across the eastern United States, from southern Canada south to Florida and west to the Great Plains. It inhabits dense shrubby undergrowth: forest edges with thick brush, overgrown fields, bramble thickets, woodland clearings, suburban gardens with dense native plantings, and pine-oak barrens. It is a permanent resident in the South and a summer resident in the North, with northern populations moving south in winter. It is closely associated with areas of deep leaf litter beneath dense low vegetation.

DIET & FEEDING

- White millet and cracked corn scattered on the ground
- Black-oil sunflower seeds picked up from beneath feeders
- Wild weed seeds and grass seeds scratched from leaf litter
- Wild berries: blackberry, blueberry, serviceberry, pokeweed
- Insects, spiders, millipedes, and earthworms
- Acorns and small nuts broken open on the ground

BEHAVIOR & SOCIAL LIFE

The Eastern Towhee's most characteristic behavior is its vigorous double-scratch foraging technique — a powerful two-footed backward kick that hurls leaf litter aside to expose seeds and invertebrates hidden beneath. This scratching can be heard from a considerable distance and is often the first indication that a towhee is nearby. Towhees are mostly solitary outside the breeding season, spending their time in dense undergrowth where they are difficult to observe. Males are territorial and persistent singers, often delivering their song from the top of a tall shrub — one of the few occasions when this secretive species is easily visible.

NESTING & BREEDING

Eastern Towhees nest on or very near the ground, placing their bulky cup nest of leaves, bark strips, and grasses in a shallow depression concealed beneath dense shrubs or in a grass clump. The female builds the nest and incubates 3–4 creamy white eggs with reddish-brown spots for 12–13 days. Both parents feed the nestlings, which fledge at 10–12 days. The species raises 2–3 broods per season. Eastern Towhees are frequent cowbird hosts, and their ground-level nests suffer high predation rates.

SONG & COMMUNICATION

The Eastern Towhee's song is one of the most recognizable in eastern North America — a clear, emphatic "drink-your-teeeeee" with the final note a long, buzzy trill rising in pitch. Males sing persistently from spring through midsummer. The call is a sharp, rising "chewink" or "towhee" — the source of both the common name and its former name, "Rufous-sided Towhee." A soft "mew" contact call is given in dense cover.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining
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The Eastern Towhee population is estimated at 26 million individuals but has declined by roughly 50% since 1970 — one of the steeper declines among common eastern songbirds. The primary cause is the loss of dense shrubby edge habitat as forests mature and close canopy, eliminating the open undergrowth this species requires. Maintaining brush piles, bramble patches, and dense native shrubs is the most effective way to support towhees in the garden.

FEEDER TIPS

- Scatter white millet and cracked corn on bare ground beneath dense shrubs
- A brush pile near the feeding area is essential — towhees need escape cover
- They rarely use elevated feeders — ground feeding is their only mode
- Plant native blackberries, serviceberries, and viburnums for natural food
- Listen for vigorous leaf-scratching sounds in thickets — the first sign of a towhee

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head & back (male)	Glossy black
Breast sides & flanks	Rich rufous-orange
Belly	White — clean and contrasting
Tail corners	White spots — flash in flight
Female — head & back	Warm brown replacing black
Eye	Red — striking in both sexes

COLOR ME!

Eastern Towhee • *Pipilo erythrophthalmus* • Module 24



MODULE 25 / 40

Rose-breasted Grosbeak

*Pheucticus ludovicianus***QUICK STATS**

Length	7.1 – 8.3 in (18 – 21 cm)
Wingspan	11.4 – 13.0 in (29 – 33 cm)
Weight	1.4 – 1.7 oz (39 – 49 g)
Lifespan	Up to 24 years (avg. 7 in wild)

PRESENTATION

The Rose-breasted Grosbeak is one of the most stunning birds to appear at eastern feeders — the male's combination of jet-black head, bright white body, and brilliant rose-red triangular breast patch is one of the most striking plumage patterns in North American ornithology. Its arrival at sunflower feeders in early May, coinciding with peak spring migration, is one of the most eagerly anticipated events of the backyard birding year. Adding to its appeal, the male sings a rich, melodious song described as a "robin who has had singing lessons" — flowing phrases delivered with unusual sweetness and complexity. Females, by contrast, resemble large streaked sparrows and are often overlooked.

HABITAT & RANGE

The Rose-breasted Grosbeak breeds across southern Canada and the northeastern United States, from British Columbia east to Nova Scotia and south through the Appalachians and Great Plains. It is a long-distance migrant, wintering from Mexico through Central America to northern South America. It inhabits mature deciduous and mixed forests, forest edges, riparian woodlands, orchards, and suburban gardens with large trees. It passes through a broad migratory corridor across the eastern and central United States each spring and fall.

DIET & FEEDING

- Black-oil sunflower seeds — the top feeder food, strongly preferred
- Sunflower chips and safflower seeds
- Wild berries and small fruits during migration and on wintering grounds
- Insects and larvae — the bulk of the summer breeding diet
- Weed seeds and wild grain in migration
- Flower buds and nectar occasionally in spring

BEHAVIOR & SOCIAL LIFE

Rose-breasted Grosbeaks are somewhat shy and deliberate at feeders, typically visiting in the early morning and late afternoon. Males are territorial on the breeding grounds but relatively tolerant of other species. They are powerful seed-crackers — their large, conical bill can split open hard seeds with ease. During migration they may appear at feeders in small groups of 5–10 birds, sometimes including both sexes and immature males with pinkish-orange breast washes. They spend most of their time in the mid-to-upper canopy and may be difficult to observe despite their loud song.

NESTING & BREEDING

Rose-breasted Grosbeaks build a loosely constructed, somewhat flimsy cup nest of twigs and plant stems in a tree fork or dense shrub, typically 5–20 feet above the ground. Clutches of 3–5 pale blue-green eggs with brown spots are incubated by both parents for 13–14 days — a rare behavior in songbirds. Both parents share equally in brooding and feeding the nestlings, which fledge at 9–12 days. Males sometimes sing softly while incubating. One brood per season.

SONG & COMMUNICATION

The Rose-breasted Grosbeak's song is a rich, flowing series of melodious whistled phrases — sweeter and more polished than an American Robin's, delivered with a liquid quality that is instantly recognizable once learned. Both sexes sing — the female's song is softer but similar in structure. The call is a distinctive, sharp "squeaky sneaker" sound — a useful identification clue in migration when birds pass overhead.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining
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The Rose-breasted Grosbeak population is estimated at 4.1 million individuals and has declined by approximately 35% since 1970. Loss of mature deciduous forest on both breeding and wintering grounds is the primary threat. Window collisions during migration are a significant additional mortality source. Planting large native trees and maintaining sunflower feeders during the May migration window are meaningful ways to support this species.

FEEDER TIPS

- Stock sunflower feeders by late April — males arrive in early May
- A large hopper or platform feeder with sunflower seeds is ideal
- Rose-breasted Grosbeaks visit feeders most actively at dawn and dusk
- Watch for females — they look like large, streaky sparrows with a big pale bill
- Apply window decals in May — this species is highly vulnerable to glass strikes

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head (male)	Glossy black — hood
Breast patch (male)	Rose-red triangle — key field mark
Back & wings (male)	Black with white wing patches
Belly (male)	White
Female — overall	Brown, streaked; white supercilium
Bill	Large, conical, pale gray

COLOR ME!

Rose-breasted Grosbeak • *Pheucticus ludovicianus* • Module 25



MODULE 26 / 40

Black-capped Chickadee

*Poecile atricapillus***QUICK STATS**

Length	4.7 – 5.9 in (12 – 15 cm)
Wingspan	6.3 – 8.3 in (16 – 21 cm)
Weight	0.3 – 0.5 oz (9 – 14 g)
Lifespan	Up to 12 years (avg. 2–3 in wild)

PRESENTATION

The Black-capped Chickadee is arguably the most beloved feeder bird in North America — a tiny, round, endlessly cheerful bird whose bold black cap, crisp white cheeks, and fearless curiosity have made it a symbol of winter resilience and backyard birding joy. It is one of the few wild birds that can be trained to take seeds from an outstretched hand, and it often does so with remarkable boldness. Despite its tiny size — barely heavier than two nickels — it is superbly adapted to survive the harshest northern winters, capable of lowering its body temperature at night to conserve energy and maintaining thousands of cached food items whose locations it can remember for weeks.

HABITAT & RANGE

The Black-capped Chickadee is found year-round across the northern United States and most of Canada, from Alaska east to Newfoundland and south through the northern tier of states, the Appalachians, and the Pacific Northwest. It inhabits deciduous and mixed forests, forest edges, suburban parks, and wooded gardens wherever there are trees large enough to provide nesting cavities and foraging opportunities. It is a permanent resident throughout its range, though it may make short irruptive movements in years when northern food supplies fail.

DIET & FEEDING

- Black-oil sunflower seeds — the absolute top choice at feeders
- Sunflower chips and safflower seeds
- Suet cakes — especially important in cold winter months
- Peanut pieces and whole peanuts
- Nyjer seed — taken occasionally
- Insects, spiders, and their eggs gleaned from bark — dominant summer food

BEHAVIOR & SOCIAL LIFE

Black-capped Chickadees live in stable winter flocks of 6–10 birds with a fixed dominance hierarchy. The dominant pair leads the flock and gets priority access to the best food. Subordinate birds are pushed to the flock's periphery, where predator risk is higher. Flock members communicate constantly with a complex vocabulary of calls. They are among the most important "nuclear species" in winter mixed-species flocks — nuthatches, Downy Woodpeckers, and kinglets follow chickadee flocks and respond to their alarm calls. Chickadees are extraordinary food cachers, hiding thousands of seeds in bark crevices, leaves, and soil, and relocating each cache within weeks.

NESTING & BREEDING

Black-capped Chickadees excavate their own nest cavities in soft, rotting wood — a capability unique among small songbirds. They also readily use nest boxes. The female builds a soft cup nest inside the cavity using moss, plant fibers, and a thick lining of mammal fur and plant down. Clutches of 6–8 white eggs with reddish-brown spots are incubated by the female for 12–13 days. Both parents feed the nestlings, which fledge at 16 days. One brood per season.

SONG & COMMUNICATION

The Black-capped Chickadee has one of the most complex vocal systems of any North American bird. Its familiar "chick-a-dee-dee-dee" call varies in the number of "dee" notes to encode the urgency of predator threats — more "dees" mean a more dangerous predator nearby. The male's clear, two-note whistled "fee-bee" song is one of the first sounds of late winter, often heard on warm days in January or February.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The Black-capped Chickadee population is estimated at 41 million individuals and is broadly stable. It is one of the most studied birds in North America, with long-term research revealing remarkable cognitive abilities including episodic-like memory and the ability to expand the hippocampus (the brain's memory center) in autumn to accommodate increased caching demands. It benefits strongly from backyard feeders and nest box programs.

FEEDER TIPS

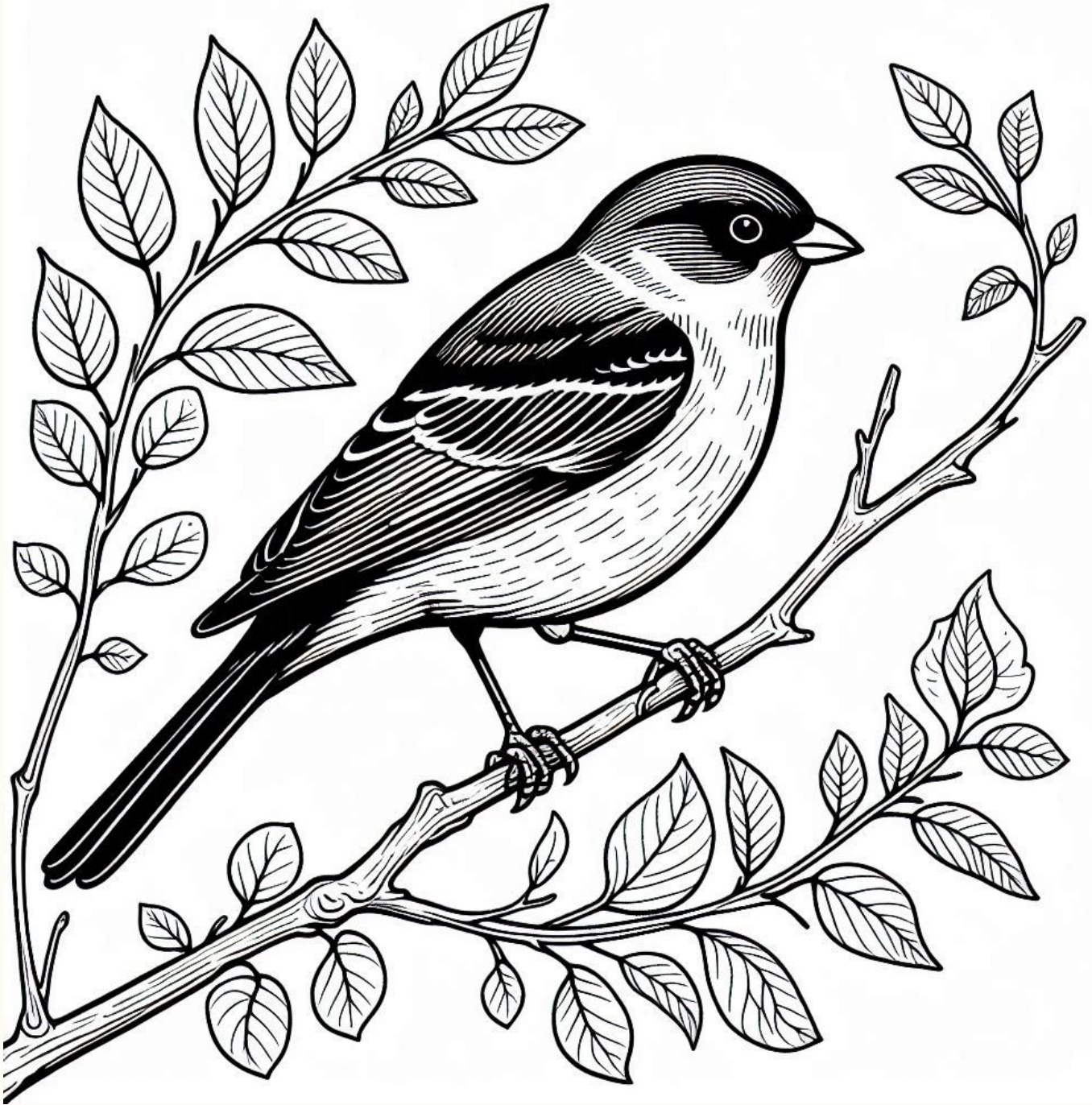
- Black-oil sunflower seeds in any feeder style — chickadees use all types
- They cache seeds — expect rapid, repeated visits as they store for winter
- Hand-feeding is achievable: stand still near the feeder with seeds in palm
- Install a nest box (1.25-inch entrance) on a tree or post — they will use it
- Suet in a cage feeder is especially valued during cold snaps below freezing

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Cap	Glossy black
Bib	Black — extends to upper breast
Cheeks	Bright white
Back & wings	Gray — soft, medium tone
Breast & belly	White to pale buff on flanks
Bill	Short, stout, black

COLOR ME!

Black-capped Chickadee • *Poecile atricapillus* • Module 26



MODULE 27 / 40

Eastern Bluebird

*Sialia sialis***QUICK STATS**

Length	6.3 – 8.3 in (16 – 21 cm)
Wingspan	9.8 – 12.6 in (25 – 32 cm)
Weight	1.0 – 1.1 oz (28 – 32 g)
Lifespan	Up to 10 years (avg. 2 in wild)

PRESENTATION

The Eastern Bluebird is one of the most beloved birds in North America — a small thrush whose brilliant royal-blue back, warm chestnut breast, and pure white belly create one of the most beautiful color combinations in the natural world. Henry David Thoreau wrote that "the bluebird carries the sky on his back," a description that remains apt. Once in severe decline due to competition from introduced House Sparrows and European Starlings for nest cavities, the Eastern Bluebird staged one of the most dramatic conservation comebacks in North American history — driven almost entirely by the efforts of volunteers erecting nest box trails across the continent.

HABITAT & RANGE

The Eastern Bluebird is found year-round across the eastern United States, from the Great Plains east to the Atlantic coast, and from southern Canada south to Nicaragua. Northern populations are partially migratory, moving south in winter. It inhabits open country with scattered trees: orchards, golf courses, roadsides, parks, open woodlands, pastures with fence posts, and suburban gardens with short grass and nearby trees. It is a cavity nester that depends entirely on pre-existing holes in trees or nest boxes.

DIET & FEEDING

- Live or dried mealworms — the single most effective food offering
- Wild berries: holly, dogwood, viburnum, pokeweed, Virginia creeper
- Insects and earthworms — the dominant food in warm months
- Bluebird-specific feeders with mealworm dishes
- Small fruits: serviceberry, hackberry, wild grape
- Rarely visits seed feeders — mealworms are essential

BEHAVIOR & SOCIAL LIFE

Eastern Bluebirds are gentle, sociable birds that form family groups through the breeding season, with offspring from earlier broods sometimes helping feed their younger siblings. In winter they gather in loose flocks of 10–20 birds that roam open country in search of berry crops. They hunt insects by perching on a low branch or fence post, scanning the ground below, then dropping to capture prey — a hunting style called "drop-hunting" or "flycatching from a perch." They communicate with each other constantly using soft, melodious warbling calls that carry a quality of gentle contentment.

NESTING & BREEDING

Eastern Bluebirds are cavity nesters, using old woodpecker holes and nest boxes. The female builds a neat cup of grasses and pine needles inside the cavity. Clutches of 4–6 pale blue (or occasionally white) eggs are incubated by the female for 13–16 days. Both parents and sometimes helper birds feed the nestlings, which fledge at 17–21 days. Pairs raise 2–3 broods per season. Aggressive monitoring and eviction of House Sparrows from nest boxes is essential for bluebird nesting success.

SONG & COMMUNICATION

The Eastern Bluebird's song is a soft, rich, melodious warble — a series of flowing phrases with a gentle, liquid quality. It is one of the most pleasant sounds of early spring in eastern North America, beginning on warm late-winter days. The call is a distinctive, soft "tru-ly" or "chur-wi" whistle that is easily recognized once learned. Males sing persistently from exposed perches throughout the breeding season.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Increasing
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The Eastern Bluebird population is estimated at 22 million individuals and has increased dramatically since the 1970s, rebounding from severe mid-20th-century declines. This recovery is one of the great citizen science success stories — driven by thousands of volunteers maintaining nest box trails across the continent. Installing and actively managing nest boxes remains the single most impactful action any individual can take for this species.

FEEDER TIPS

- Offer live or dried mealworms in a shallow dish feeder near open grass
- Use a bluebird-specific feeder with a small entry hole to exclude starlings
- Install a nest box (1.5-inch entrance) on a post in open area, 5 ft high
- Monitor boxes weekly — evict House Sparrow nests immediately
- Plant native berry-producing shrubs: holly, dogwood, and serviceberry

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Back & wings (male)	Brilliant royal blue
Breast & flanks (male)	Rich chestnut-orange
Belly (male)	White
Female — back	Blue-gray with blue tinge on wings
Female — breast	Pale orange-buff, washed
Bill	Short, slightly curved, dark

COLOR ME!

Eastern Bluebird • *Sialia sialis* • Module 27



MODULE 28 / 40

American Robin

*Turdus migratorius***QUICK STATS**

Length	7.9 – 11.0 in (20 – 28 cm)
Wingspan	12.2 – 15.8 in (31 – 40 cm)
Weight	2.7 – 3.0 oz (77 – 85 g)
Lifespan	Up to 14 years (avg. 2 in wild)

PRESENTATION

The American Robin is perhaps the most universally recognized bird in North America — a large, familiar thrush whose brick-red breast, dark back, yellow bill, and bold white eye ring are known to virtually every North American from childhood. It is the state bird of Connecticut, Michigan, and Wisconsin, and its cheerful caroling song — a rich, flowing "cheerily, cheer up, cheer up, cheerily, cheer up" — is one of the quintessential sounds of spring. The American Robin is often cited as the first sign of spring, though many populations actually winter in the northern states, gathering in large nomadic flocks that feed on berries.

HABITAT & RANGE

The American Robin breeds across virtually all of North America, from Alaska and northern Canada south through Mexico. It is one of the most widespread and abundant birds on the continent. It inhabits an extraordinary range of environments: forest edges, open woodlands, suburban lawns, parks, golf courses, farmland, alpine meadows, and tundra edges. It is a year-round resident in most of the United States, though northern populations are partially migratory. In winter, flocks gather wherever fruiting trees are abundant.

DIET & FEEDING

- Earthworms — pulled from lawns, the iconic foraging image
- Wild berries: holly, crabapple, mountain ash, juniper, dogwood
- Insects and larvae during the breeding season
- Small fruits: cherry, grape, serviceberry, hackberry
- Occasionally visits platform feeders for mealworms or fruit
- Robins rarely use seed feeders — berries and worms are essential

BEHAVIOR & SOCIAL LIFE

American Robins are highly territorial during the breeding season but become intensely gregarious in winter, gathering in nomadic flocks of hundreds to thousands of birds that roam the landscape in search of fruiting trees. Their famous earthworm-hunting behavior — running a few steps, stopping abruptly, cocking the head, then plunging the bill into the ground — is based primarily on sight, not sound, despite popular belief. Robins are among the earliest morning singers, beginning their caroling well before dawn — a behavior driven by the low light levels at which their large eyes can detect prey before most other birds can forage.

NESTING & BREEDING

The American Robin builds one of the most recognizable nests in North America — a robust cup of grasses and mud, often placed on a building ledge, porch light, or tree branch 5–25 feet above the ground. The mud forms a hard, durable outer cup. Clutches of 3–5 vivid blue eggs ("robin's egg blue") are incubated by the female for 12–14 days. Both parents feed the nestlings, which fledge at 14–16 days. Pairs raise 2–3 broods per season, reusing or building new nests.

SONG & COMMUNICATION

The American Robin's song is a rich, flowing carol of rising and falling phrases — "cheerily, cheer up, cheer up, cheerily, cheer up" — delivered in long, continuous bouts from a treetop. It is one of the first bird songs heard at dawn and often one of the last at dusk. The call is a sharp "tut-tut-tut" or "whinny" alarm, instantly recognizable once learned. Both sexes produce calls, but only males deliver the full caroling song.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The American Robin population is estimated at 300 million individuals — one of the most abundant birds in North America — and is broadly stable. It has benefited greatly from the spread of suburban lawns and ornamental berry trees. Pesticide use on lawns poses a significant risk, as robins consume large quantities of earthworms that accumulate chemical residues. Avoiding lawn pesticides is one of the most meaningful steps a homeowner can take for robins.

FEEDER TIPS

- Plant native fruiting trees: crabapple, hawthorn, holly, and serviceberry
- Offer mealworms in a shallow platform dish — robins will come for them
- Maintain a lawn area with moist soil for earthworm foraging
- A shallow ground-level birdbath is strongly attractive to robins
- Robins nest on ledges and porches — a simple wooden shelf can host a nest

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head (male)	Dark charcoal-black
Back & wings	Gray-brown
Breast & belly	Rich brick-orange red
Throat	White with dark streaks
Eye ring	Bold white — key field mark
Bill	Yellow-orange, fairly long

COLOR ME!

American Robin • *Turdus migratorius* • Module 28



MODULE 29 / 40

Hermit Thrush

*Catharus guttatus***QUICK STATS**

Length	5.5 – 7.1 in (14 – 18 cm)
Wingspan	9.8 – 11.4 in (25 – 29 cm)
Weight	0.8 – 1.3 oz (23 – 37 g)
Lifespan	Up to 11 years (avg. 4 in wild)

PRESENTATION

The Hermit Thrush is widely considered to produce the most beautiful song of any North American bird — a series of ethereal, flute-like phrases of haunting purity, each beginning on a long introductory note and cascading through a series of spiraling harmonics. Walt Whitman immortalized it in "When Lilacs Last in the Dooryard Bloom'd," and it is the state bird of Vermont. Despite its modest, spotted brown plumage, the Hermit Thrush is instantly distinguished from similar thrushes by its habit of slowly raising and lowering its bright rufous tail — a unique, meditative gesture that no other North American thrush performs.

HABITAT & RANGE

The Hermit Thrush breeds across the boreal forests of Canada and Alaska, and in montane coniferous forests south through the Appalachians, Rockies, and Sierra Nevada. It is the only spotted thrush to winter widely in the United States — a critical identification clue in winter when other *Catharus* thrushes have departed for the tropics. In winter it frequents forest understories, dense shrubby thickets, and wooded gardens with berry-producing shrubs, from the Pacific coast to the Atlantic seaboard.

DIET & FEEDING

- Wild berries: holly, dogwood, mistletoe, juniper, serviceberry
- Insects and earthworms gleaned from leaf litter
- Small fruits: hackberry, pokeweed, wild grape
- Spiders and small invertebrates from the forest floor
- Mealworms offered in a low platform or ground dish
- Rarely visits seed feeders — prefers natural food in dense cover

BEHAVIOR & SOCIAL LIFE

The Hermit Thrush earns its name from its solitary, retiring nature — it is almost always seen alone, quietly foraging through leaf litter or perching in dense understory vegetation. Its tail-raising behavior is one of the most distinctive field marks of any North American bird: the bird slowly lifts its rufous tail to a near-vertical position, holds it briefly, then lowers it gradually — a behavior repeated constantly while the bird is perched. On the breeding grounds males are territorial, singing from high perches at dawn and dusk. In winter it is among the most solitary of thrushes, defending individual berry-producing shrubs against all comers.

NESTING & BREEDING

The Hermit Thrush nests on or near the ground in dense boreal or montane forest. The female builds a bulky cup nest of leaves, grass, and bark, well concealed beneath a low conifer or shrub. Clutches of 3–4 pale blue eggs are incubated by the female for 11–13 days. Both parents feed the nestlings, which fledge at 10–15 days. Pairs raise 1–2 broods per season. The species is generally a poor cowbird host due to the remoteness of its nesting habitat.

SONG & COMMUNICATION

The Hermit Thrush's song stands alone in North American ornithology for its otherworldly beauty — a long, clear introductory whistle followed by a spiraling cascade of pure, harmonically rich phrases, each at a different pitch. No two phrases are alike. The song carries through dense forest with extraordinary clarity. The call is a soft, rising "veer" or a sharp "chuck" alarm note. Song is heard primarily at dawn and dusk on the breeding grounds.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The Hermit Thrush population is estimated at 22 million individuals and is broadly stable. As the only *Catharus* thrush to winter in North America, it is less threatened by tropical deforestation than its relatives. Climate change poses a long-term threat by shifting boreal forest composition. Window collisions during migration are a significant mortality factor, as this species migrates at night.

FEEDER TIPS

- Plant native holly, dogwood, and serviceberry — essential winter food
- Offer mealworms in a low, sheltered dish near dense shrubs
- A ground-level birdbath near dense cover is highly attractive
- Hermit Thrushes are secretive — watch for them creeping along the ground in thickets
- The key field mark: watch for the slow, deliberate tail-raising behavior

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Back & head	Warm olive-brown
Rump & tail	Bright rufous — key field mark
Breast	White with bold dark spots
Flanks	Buff-gray wash
Eye ring	Pale white — distinct
Bill	Slender, slightly curved, dark

COLOR ME!

Hermit Thrush • *Catharus guttatus* • Module 29



MODULE 30 / 40

Northern Mockingbird

Mimus polyglottos

QUICK STATS

Length	8.3 – 10.2 in (21 – 26 cm)
Wingspan	12.2 – 13.8 in (31 – 35 cm)
Weight	1.4 – 2.0 oz (40 – 58 g)
Lifespan	Up to 14 years (avg. 8 in wild)

PRESENTATION

The Northern Mockingbird is one of the most vocally gifted birds in the world — a medium-sized, gray-and-white songbird capable of learning and performing the songs of dozens of other species, sometimes singing continuously for hours without repeating itself. Its scientific name, *Mimus polyglottos*, means "many-tongued mimic," and it earns that title fully: a single male may have a repertoire of over 200 distinct song types. It is the state bird of five U.S. states — Arkansas, Florida, Mississippi, Tennessee, and Texas — and its bold, assertive personality makes it one of the most conspicuous birds in suburban and urban America.

HABITAT & RANGE

The Northern Mockingbird is a year-round resident across the southern half of the United States, from California to the Atlantic coast, and has expanded its range northward over recent decades, now regularly breeding in southern Canada. It thrives in open and semi-open habitats: suburban gardens, parks, hedgerows, roadsides, forest edges, and agricultural areas with scattered shrubs and trees. It shows a particular affinity for fruiting shrubs such as holly, pyracantha, and multiflora rose.

DIET & FEEDING

- Wild berries: holly, pyracantha, pokeweed, multiflora rose, elderberry
- Insects and earthworms — dominant in warm months
- Small fruits: crabapple, grape, hackberry
- Suet and mealworms occasionally at feeders
- Lizards and small frogs taken occasionally
- Flower nectar and fruit juice probed directly in summer

BEHAVIOR & SOCIAL LIFE

Northern Mockingbirds are famously bold and aggressive, fearlessly attacking birds many times their size — hawks, crows, and even cats and dogs — that venture near their nesting or feeding territories. They maintain separate territories in summer (for breeding) and winter (for berry supplies), and defend both vigorously. Males are among the most persistent singers in North America, sometimes singing through the night during the full moon — a behavior that can test the patience of neighboring humans. Their wing-flashing display — spreading the wings suddenly to reveal white patches — may startle insects into movement, making them easier to catch.

NESTING & BREEDING

Northern Mockingbirds build bulky cup nests of twigs and plant material, lined with grasses and rootlets, typically in a dense shrub or small tree 3–10 feet above the ground. Both parents build the nest. Clutches of 3–5 pale blue-green eggs with brown spots are incubated primarily by the female for 12–13 days. Both parents feed the nestlings, which fledge at 11–13 days. Pairs raise 2–3 broods per season and are aggressively defensive near the nest.

SONG & COMMUNICATION

The Northern Mockingbird's song is a continuous, exuberant medley of imitated and original phrases, each repeated 3–5 times before switching to the next. A male may mimic cardinals, jays, hawks, frogs, car alarms, and squeaky gates — anything it hears repeatedly. Unmated males sing most persistently, sometimes all night. The call is a sharp "chuck" or harsh "chat." Song is heard year-round in warm climates.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining slightly
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The Northern Mockingbird population is estimated at 31 million individuals but has declined by roughly 21% since 1970. Loss of dense fruiting shrub habitat through suburban development and the removal of "weedy" hedgerows and thickets are the primary drivers. Planting native fruiting shrubs — especially holly and native roses — and maintaining dense hedges are the most effective ways to support this species.

FEEDER TIPS

- Plant native holly, pyracantha, and beautyberry — far more effective than feeders
- Mockingbirds rarely visit seed feeders; offer fruit pieces or mealworms on a platform
- They aggressively defend berry shrubs — plant enough for multiple birds
- A heated birdbath in winter is strongly attractive to them
- Expect them to chase other birds away from feeders — give them their own space

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Back & head	Medium gray
Wings	Dark gray with white wing patches
Tail	Long, dark; white outer feathers
Underparts	Pale gray-white
Eye	Pale yellow — striking
Bill	Long, slightly curved, dark

COLOR ME!

Northern Mockingbird • *Mimus polyglottos* • Module 30



MODULE 31 / 40

Ruby-throated Hummingbird

Archilochus colubris

QUICK STATS

Length	2.8 – 3.5 in (7 – 9 cm)
Wingspan	3.1 – 4.3 in (8 – 11 cm)
Weight	0.1 – 0.2 oz (2.4 – 6.1 g)
Lifespan	Up to 9 years (avg. 3–5 in wild)

PRESENTATION

The Ruby-throated Hummingbird is the only hummingbird that breeds in eastern North America — a gem of iridescent emerald and ruby that seems too brilliant and too small to be real. Weighing less than a penny, it beats its wings 53 times per second in normal flight and up to 200 times per second in a courtship dive, generating the characteristic humming sound that gives the family its name. It is a physiological marvel: its heart beats up to 1,260 times per minute during flight, and it enters torpor each night — dropping its body temperature dramatically to conserve energy — yet crosses the Gulf of Mexico in a single non-stop flight of 500 miles during migration.

HABITAT & RANGE

The Ruby-throated Hummingbird breeds across eastern North America, from southern Canada south through the entire eastern United States to the Gulf Coast and into Central America. It winters from Mexico through Central America. It inhabits forest edges, gardens, orchards, and any open area with flowering plants. It shows a particular preference for tubular red and orange flowers: trumpet vine, cardinal flower, bee balm, and salvia. It arrives in the East in late April and departs by early October.

DIET & FEEDING

- Nectar — from both flowers and hummingbird feeders (4:1 water:sugar solution)
- Small insects and spiders — essential protein, especially for females and nestlings
- Tree sap from sapsucker wells — used during early spring before flowers bloom
- Trumpet vine, cardinal flower, bee balm, columbine, salvia nectar
- Gnats and small flies caught in aerial sallies
- Never use red dye in feeders — plain sugar water is safest

BEHAVIOR & SOCIAL LIFE

Ruby-throated Hummingbirds are fiercely territorial and largely solitary. Males defend feeding territories with aggressive aerial chases, driving away rival males and often females as well. Their courtship display is spectacular: the male performs pendulum-like U-shaped or figure-eight dives in front of a perched female, the rush of air through the tail feathers creating a distinctive chirping sound. Outside the breeding season individuals are independent and non-social. They have extraordinary spatial memory, returning to the exact same feeders and flowers year after year within days of their spring arrival date.

NESTING & BREEDING

The female builds a tiny, exquisite cup nest the size of a large thimble, constructed from plant down and bud scales bound together with spider silk — which allows the nest to expand as the nestlings grow. It is camouflaged with lichens on the outside and placed on a downward-sloping branch. She lays 2 white eggs the size of small jellybeans and incubates them alone for 11–16 days. She feeds the nestlings regurgitated nectar and insects. Young fledge at 18–22 days. The male plays no role in nesting.

SONG & COMMUNICATION

The Ruby-throated Hummingbird's primary sound is its wing hum — the rapid wingbeats produce a distinctive buzzing tone that varies with flight speed and behavior. Vocal calls are a rapid, chittering "chee-dit" and a series of sharp chips used in aggressive encounters. Males produce a distinctive tail-chirp during courtship dives. The species has no true song.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The Ruby-throated Hummingbird population is estimated at 34 million individuals and is broadly stable. It is one of the few migratory species whose range has benefited from the proliferation of backyard feeders and native plantings. Threats include pesticide use reducing insect prey, habitat loss on wintering grounds, and window collisions. Planting native flowers and keeping feeders clean are the most effective support actions.

FEEDER TIPS

- Mix 1 part white sugar to 4 parts water — no red dye, no honey
- Clean feeders every 2–3 days in summer heat to prevent mold and fermentation
- Put feeders out by late April in the East — males arrive before females
- Plant trumpet vine, cardinal flower, bee balm, and salvia for natural nectar
- Multiple feeders spaced apart reduce territorial aggression at feeding sites

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Back & crown (male)	<i>Iridescent emerald green</i>
Throat gorget (male)	<i>Ruby red — iridescent, angle-dependent</i>
Underparts	<i>White to pale gray</i>
Female — throat	<i>White, may show faint streaking</i>
Wings	<i>Dark — blur at 53 beats/second</i>
Bill	<i>Long, needle-straight, black</i>

COLOR ME!

Ruby-throated Hummingbird • *Archilochus colubris* • Module 31



MODULE 32 / 40

Steller's Jay

Cyanocitta stelleri

QUICK STATS

Length	11.8 – 13.4 in (30 – 34 cm)
Wingspan	17.3 in (44 cm)
Weight	3.5 – 4.9 oz (100 – 140 g)
Lifespan	Up to 16 years (avg. 6 in wild)

PRESENTATION

Steller's Jay is the western counterpart of the Blue Jay — a large, crested jay of western coniferous forests whose dramatic two-toned plumage, sooty black in front and deep cobalt blue behind, makes it one of the most striking birds at western feeders. Named after the German naturalist Georg Wilhelm Steller, who collected the first specimen in Alaska in 1741, it is the only crested jay found west of the Rocky Mountains. Bold, raucous, and highly intelligent, it is a dominant presence at feeders and campsites throughout the mountains and forests of the Pacific states, British Columbia, and the Rocky Mountain region.

HABITAT & RANGE

Steller's Jay is found year-round from southern Alaska south through British Columbia and the Pacific states to Central America, and east through the Rocky Mountains. It is a bird of coniferous and mixed forests, particularly favoring mature pine, fir, and spruce forests at mid to high elevations. In winter it descends to lower elevations and readily visits suburban feeders and campgrounds. It is a permanent resident throughout most of its range, though montane populations make seasonal elevational movements.

DIET & FEEDING

- Acorns and pine seeds — the core of the natural diet, cached for winter
- Black-oil sunflower seeds and peanuts at feeders
- Suet cakes and corn
- Insects, beetles, and grasshoppers in summer
- Eggs and nestlings of other birds — an opportunistic predator
- Berries, small fruits, and carrion scavenged from campsites

BEHAVIOR & SOCIAL LIFE

Steller's Jays are bold, assertive, and highly intelligent birds that quickly learn to associate humans with food. They are among the most dominant species at western feeders, displacing smaller birds with ease. Like other corvids, they are avid food cachers — storing acorns, pine seeds, and sunflower seeds in the ground and in bark crevices, relocating caches with remarkable accuracy weeks or months later. They are also accomplished mimics, frequently imitating the calls of Red-tailed Hawks, Osprey, and other raptors — possibly to frighten other birds away from food sources. Outside the breeding season they often gather in loose groups.

NESTING & BREEDING

Steller's Jays build large, bulky cup nests of sticks, leaves, and mud in the fork of a conifer, typically 10–40 feet above the ground. Both parents build the nest. Clutches of 3–5 pale green or blue eggs with dark spots are incubated primarily by the female for 16–18 days. Both parents feed the nestlings, which fledge at 17–19 days. One brood per season. Pairs mate for life and maintain year-round territories.

SONG & COMMUNICATION

Steller's Jay is one of the most vocal birds in western forests, producing a wide array of harsh, loud calls. The most common is a repeated "shook-shook-shook" or "shek-shek-shek" — a penetrating series that carries through dense forest. It also gives a harsh "waaaah" alarm and mimics the calls of Red-tailed Hawks, Golden Eagles, and other species with impressive accuracy. Soft, gurgling notes are used between mated pairs.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining slightly
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The Steller's Jay population is estimated at 2.8 million individuals and has shown modest declines linked to habitat loss in old-growth coniferous forests and the effects of climate change on montane ecosystems. It is highly dependent on mature forest with large nut-producing trees. Retaining old-growth conifers and offering supplemental food at western feeders support local populations.

FEEDER TIPS

- Whole peanuts and sunflower seeds on a large platform feeder work best
- Steller's Jays are dominant — place feeders for smaller birds out of sight
- They cache food — expect them to carry multiple seeds or nuts at once
- Suet in a cage near conifers is readily taken in winter
- They are bold — will approach within feet of a patient, still observer

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head, crest & back (front)	Sooty black
Body & wings	Deep cobalt blue
Fine wing barring	Pale blue — visible up close
Crest	Tall, prominent, black
Underparts	Dark blue — not as bright as back
Bill & legs	Black

COLOR ME!

Steller's Jay • *Cyanocitta stelleri* • Module 32

